

ME 310
HEAT EQUIPMENT DESIGN
SESSIONAL

PROJECT NAME
DOUBLE PIPE HEAT EXCHANGER
WITH DISK TYPE TURBULATOR AND
OUTSIDE FINS





INTRODUCTION

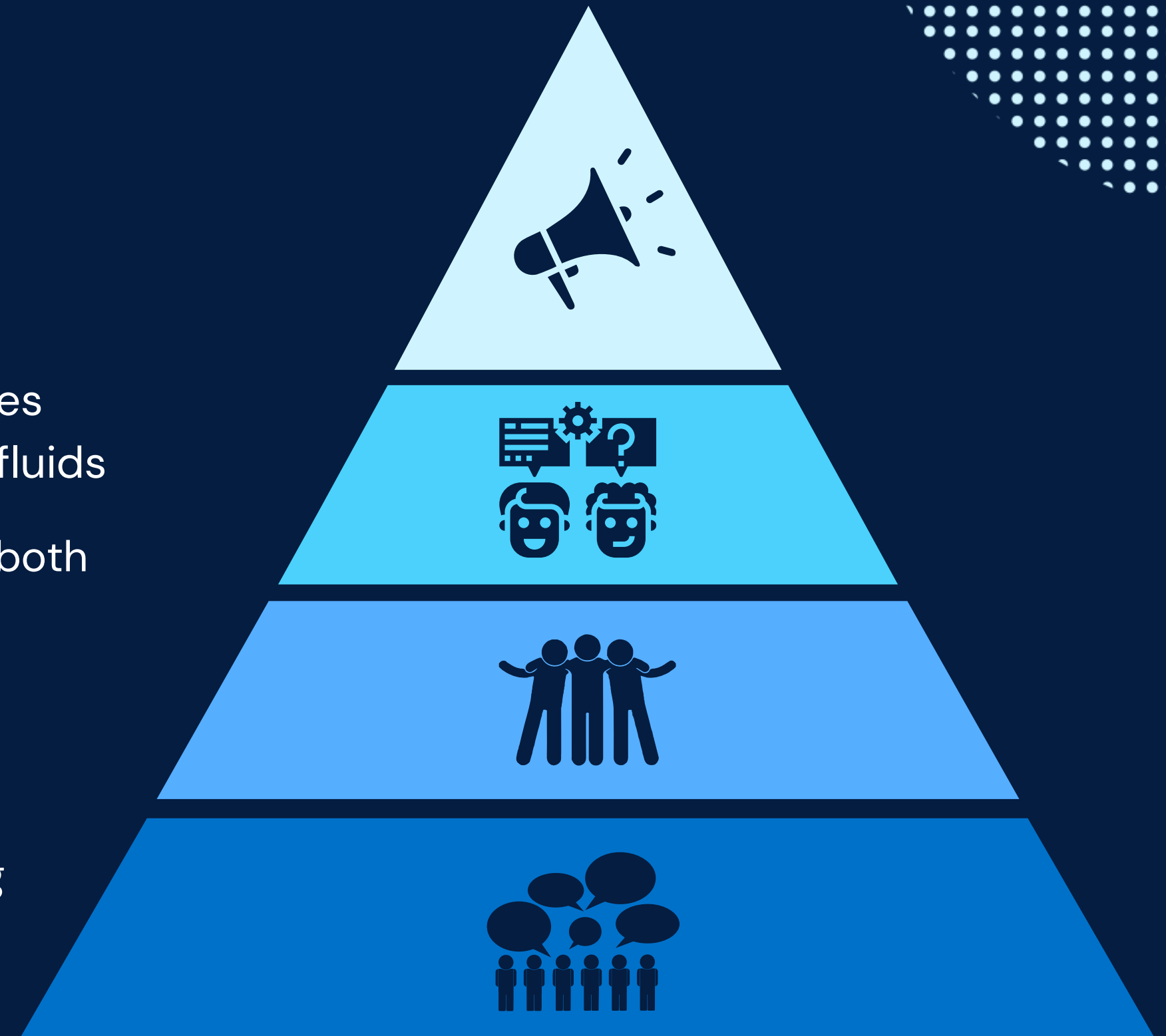
Double Pipe Heat Exchangers are used for both heating and cooling purposes. It consists of 2 concentric tubes with an annular region and a pipe region.

Hot fluid flows through one region while cold fluid flows through the other region. Heat is exchanged between these 2 fluids via conduction & convection through the pipe walls.

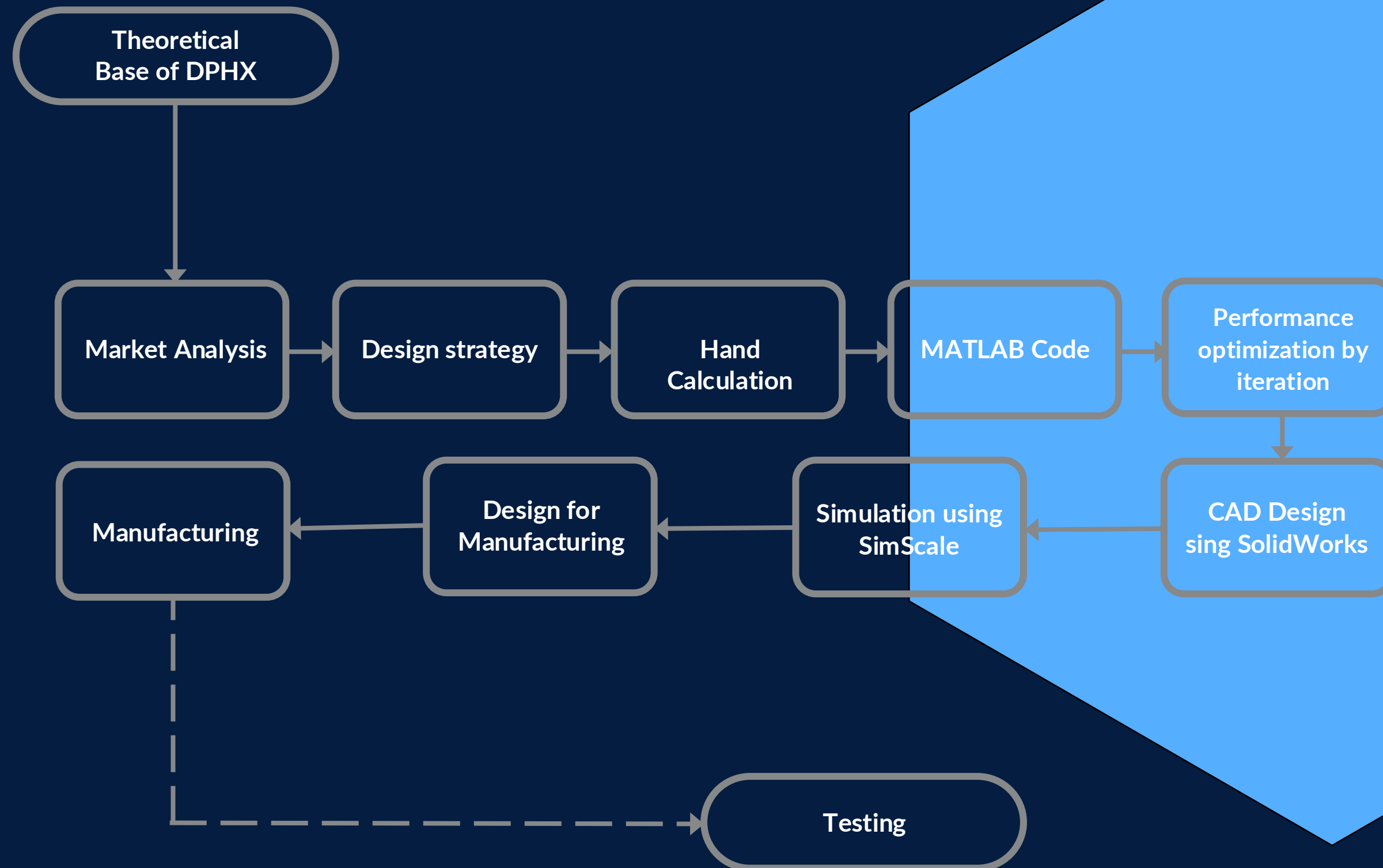
Fins and Turbulators are extra features added to increase the overall heat transfer rate between the 2 operating fluids.

OBJECTIVES

- 01 Investigate the change in temperatures along the outlets of the hot and cold fluids
- 02 Investigate the pressure drops across both the inner and outer pipes
- 03 Decide on an optimum design for the double pipe heat exchanger
- 04 Validate the performance from running MATLAB code and simulations



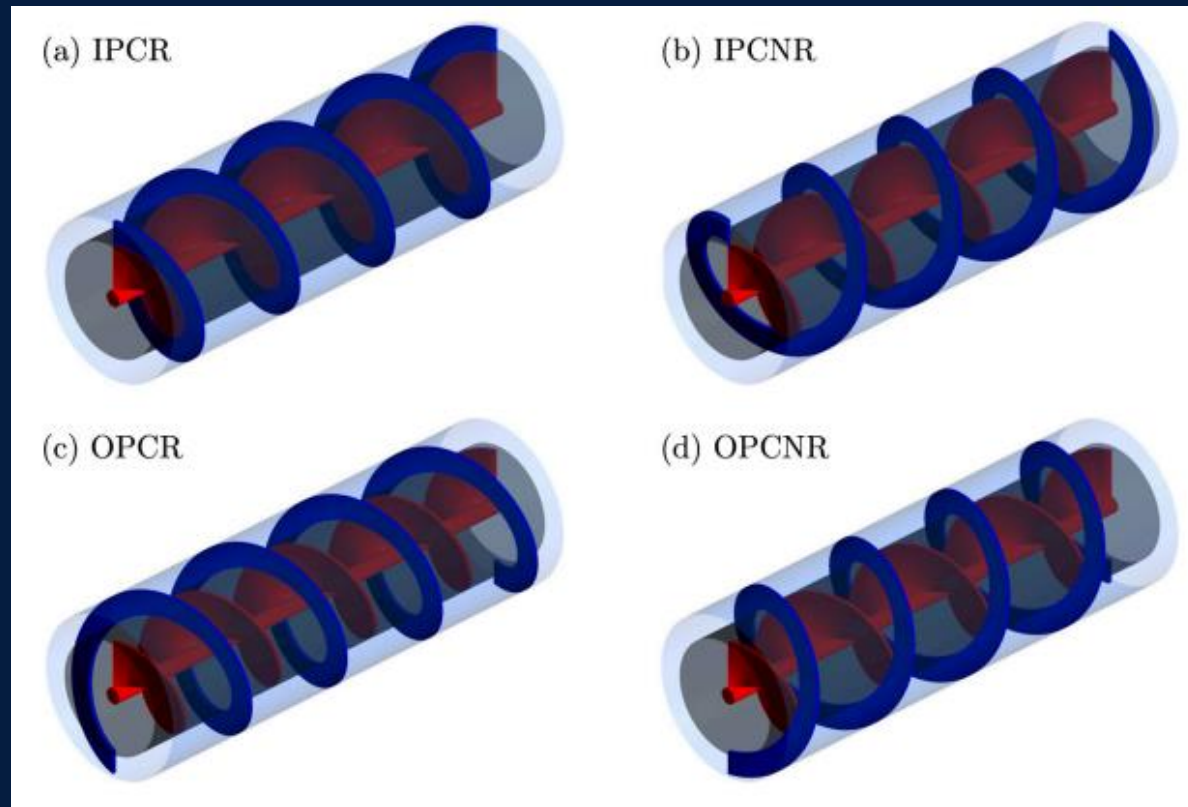
WORKFLOW



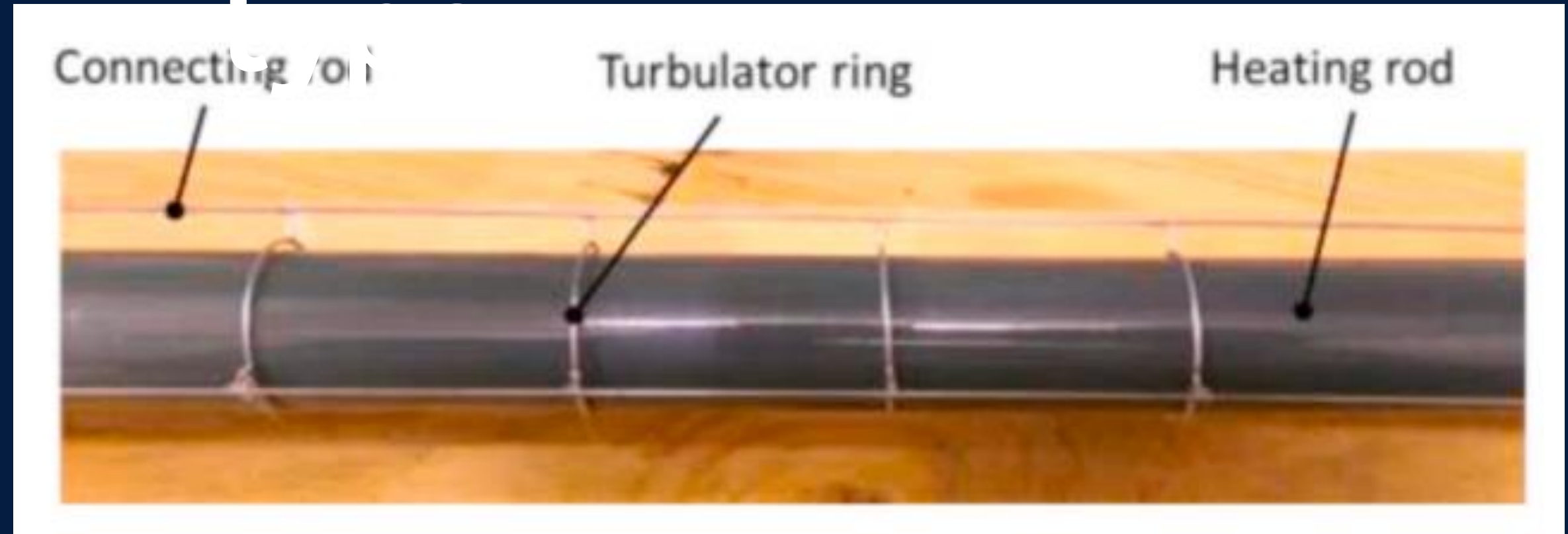
PROBLEM STATEMENT

A double pipe heat exchanger with 'Perforated shape annular turbulator' & 'Outer fins' is to be constructed in which hot water flows through annular region & cold water in inner tube. Pressure drop should be less than 1 KPa in both region.

Choosing the turbulator



Helical Type



Ring Type

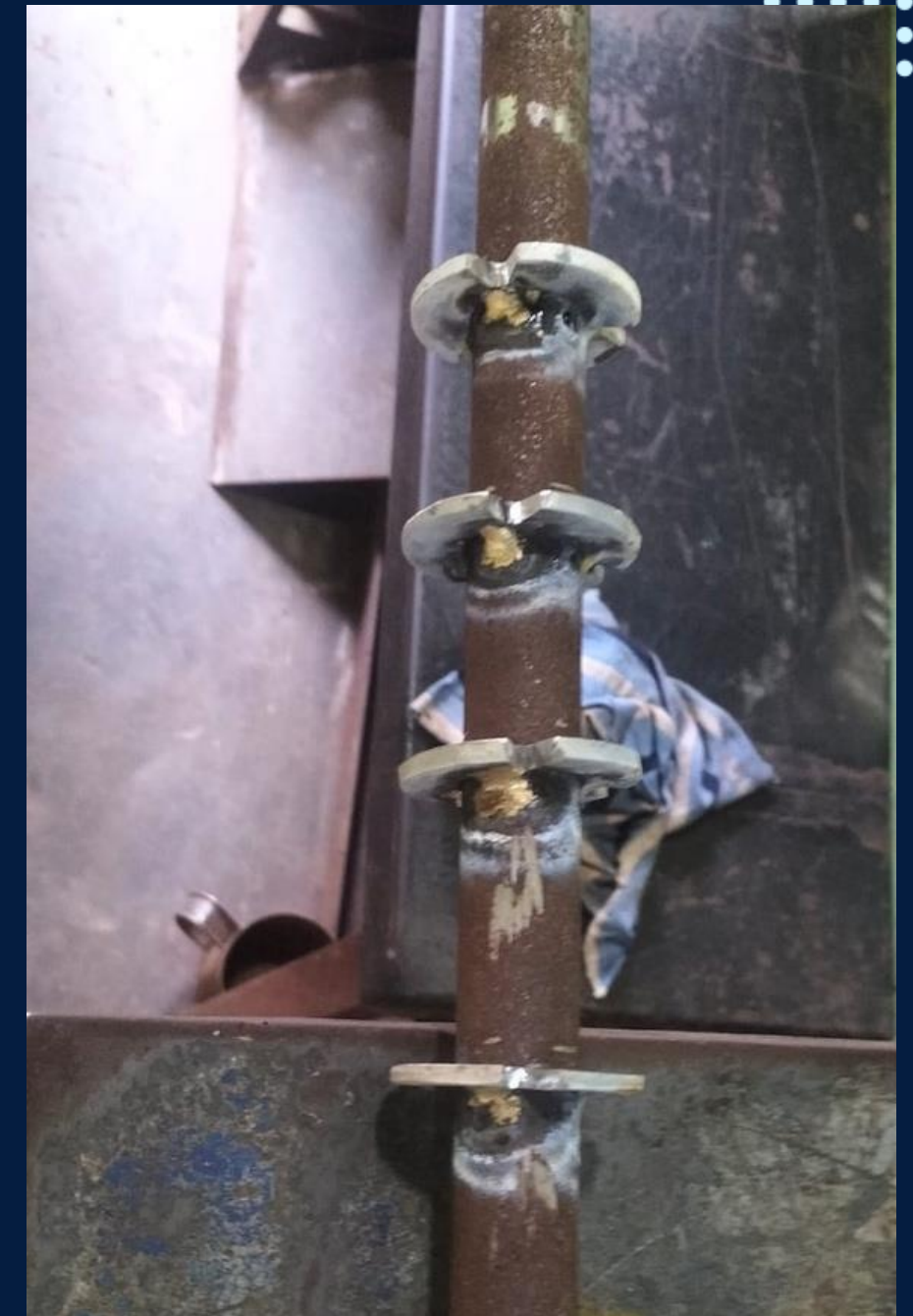


Perforated Disc

Perforated Type Turbulator

Features:

- 13 Discs made of Stainless Steel welded onto the surface of the inner pipe
- Discs are equidistant (60mm apart)
- Discs have 3 slots each for the fluid to flow
- Allows for the creation of turbulence in fluid flow, resulting in increased heat transfer



Why place fins on the outside?

- Increase the effective surface area
- Facilitate the overall heat transfer coefficient
- Facilitate the cooling down of hot fluid by natural convection



Selection of Fin Material

Available Options

- Copper($k=398\text{W/mK}$)
- Aluminium(247W/mK)
- Mild Steel(45W/mK)

Problems with Copper:

- Expensive
- Dissimilar material
- Difficult to weld

Problems with Aluminium:

- Soft(15HB)
- Very difficult to weld($MP = 660^\circ\text{C}$)
- Dissimilar metal from outer pipe

Advantages of Mild Steel as Fin:

- Melting point is between 1350-1530°C
- Easy to weld
- Similar metal as outer pipe
- Similar thermal expansion coefficients
- Fewer residual stresses developed after welding due to similar properties

INITIAL ASSUMPTIONS

Length=1 m

Inlet & Outlet
Temperature

Mass flow rate

Material &
Radial Clearance

Flow velocity

Hot water inlet temperature= 90 °C
Cold water inlet temperature =25 °C

Hot water mass flow rate 0.08 kg/s
Cold water mass flow rate 0.2 kg/s

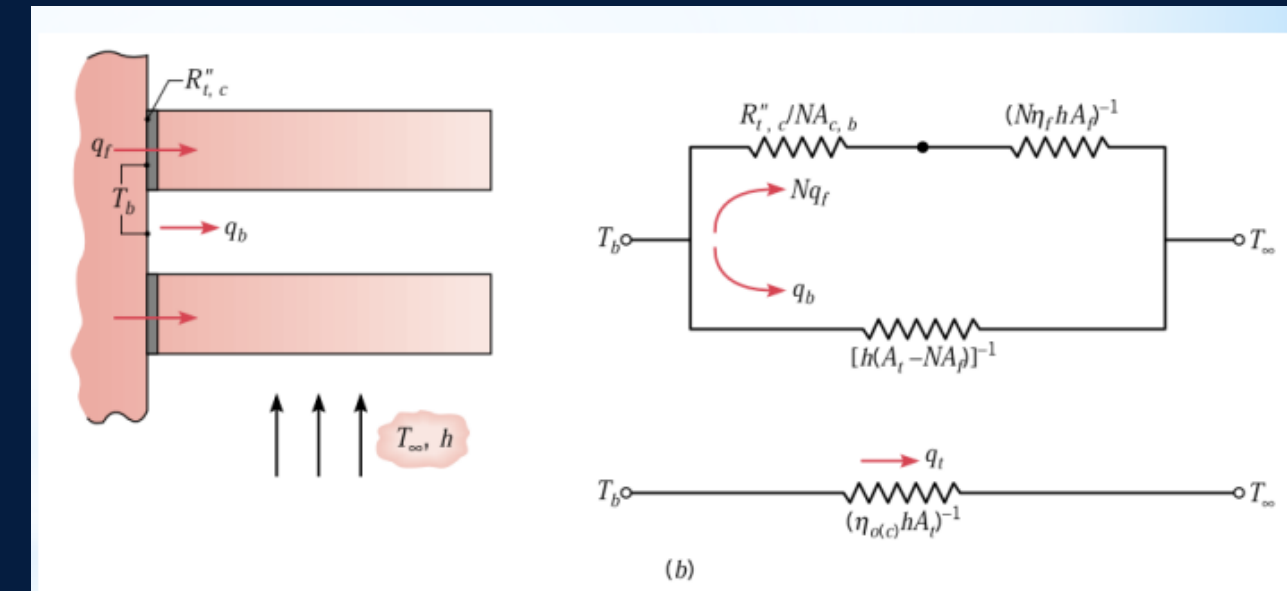
Hot Water velocity =0.13 m/s
Cold water velocity =0.27 m/s

Mild steel
Radial clearance in annular region > 0.5
inch

Pressure drop in both pipes
< 500 Pa

MATLAB CODE WORKING

- Inlet & outlet temperature, mass flow rate & velocity in both pipes are assumed.
- No other manual inputs are needed.
- Code calculates the fluid properties via linear interpolation in property tables.
- Inner pipe diameter is found from required flow area, code matches the nearest diameter from tube table & extracts the corresponding outer diameter. Same procedure is repeated for outer pipe selection.
- Reynolds, Nusselt number is calculated & heat transfer coefficient is determined.
- By NTU effectiveness method, total heat transferred is determined.
- Thermal resistance concept is used for fin analysis. Heat transfer via fins are calculated.
- Outlet temperatures were determined.
- Pressure drop is calculated.
- Many iterations were done to select sizing & specifications.



Parameters	Inner Tube	Outer Tube
Inlet Temp.(°C)	25	90
Wall Thickness(mm)	3.00	4.00
Outlet Temp(°C)	34	86.4
Mass Flow Rate(kg/s)	0.08	0.2
Velocity(m/s)	0.27	0.13
Pressure Drop(Pa)	67.48	17.48
Heat Transfer Coefficient((W/m ² K)	1574	925
Outer Diameter (inch)	1	2.375

Overall Heat Transfer Coefficient(W/m K) ²	583
Hot Water Temperature (°C)	3.60
Radial Clearance	0.523

PARAMETRIC ANALYSIS

01

Pipe mass
flow rate

02

Annular mass
flow rate

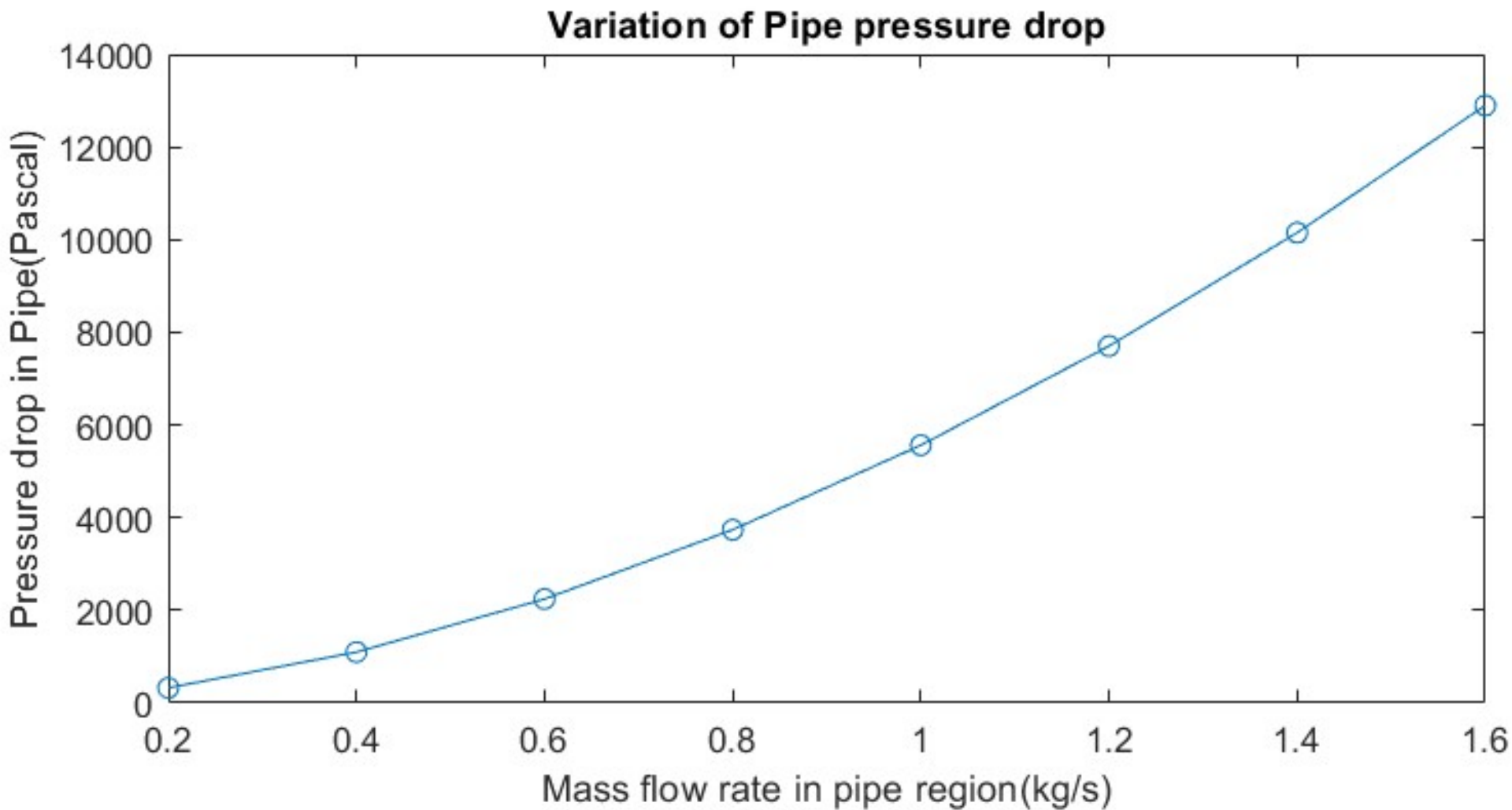
03

Length

04

Numer of
fins

INNER TUBE MASS FLOW RATE: PRESSURE DROP



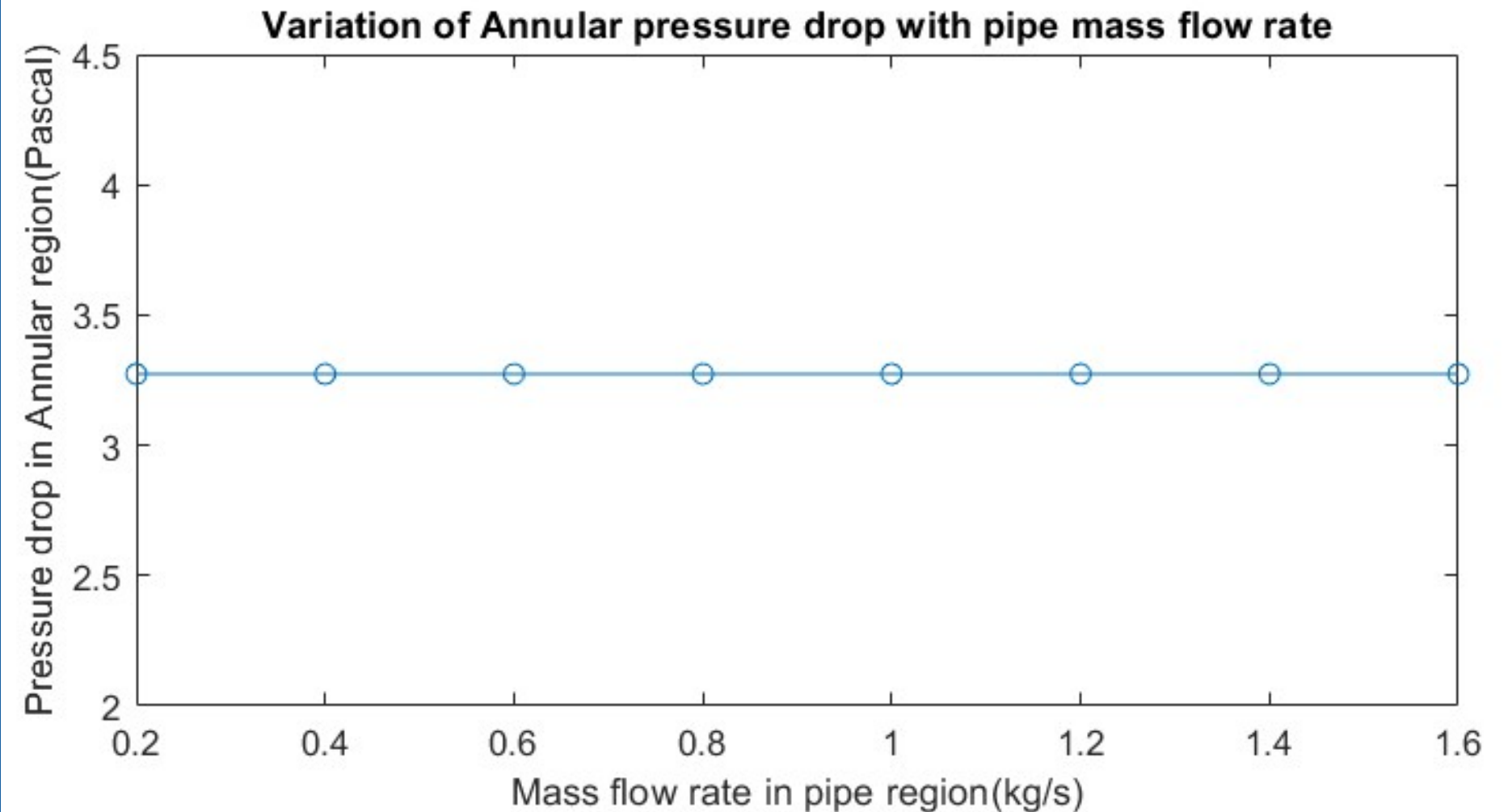
Inner tube pressure drop

Darcy-Weisbach equation:

Pipe pressure drop $\propto$ square of velocity.

Velocity $\propto$ mass flow rate.

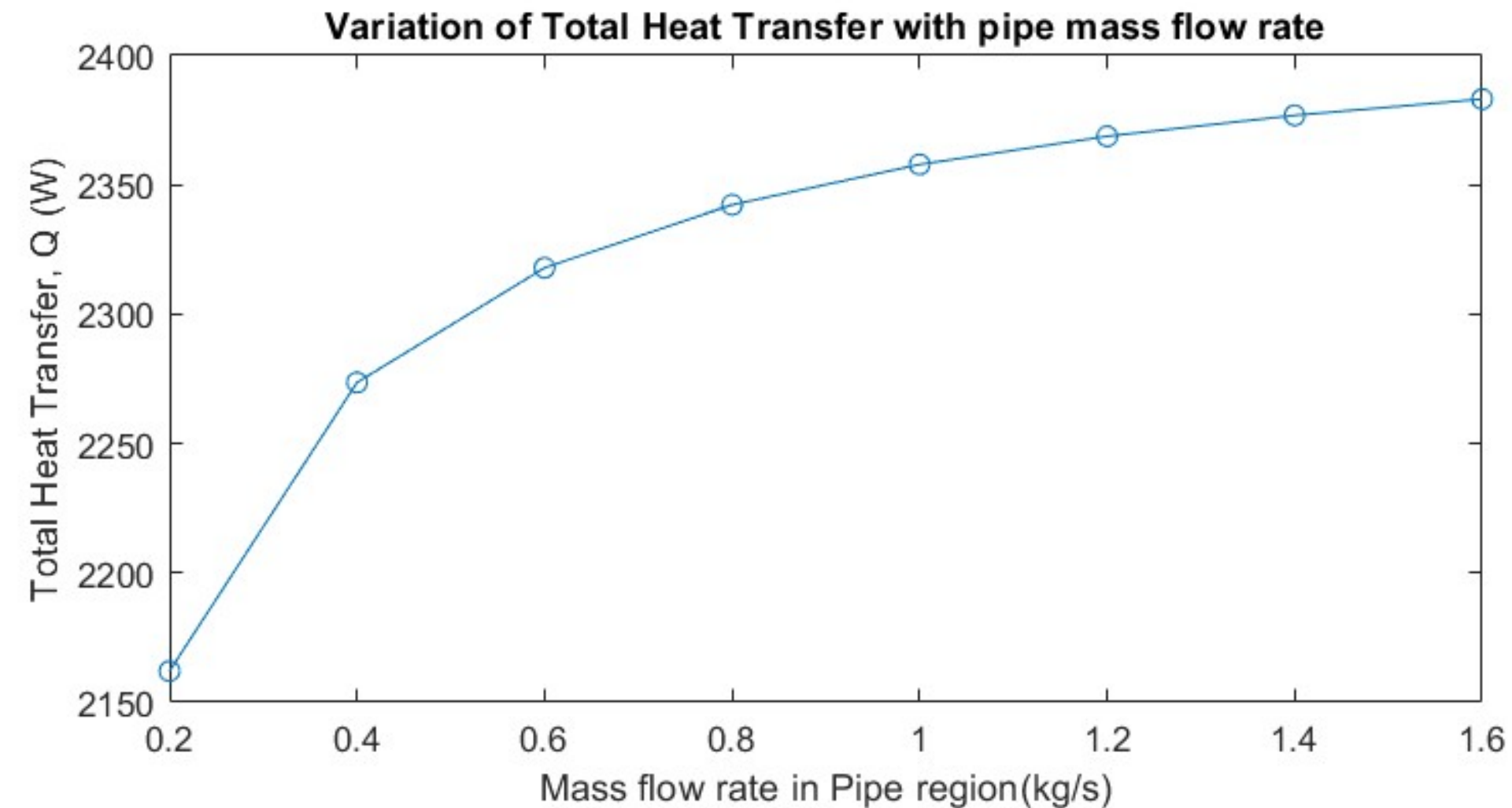
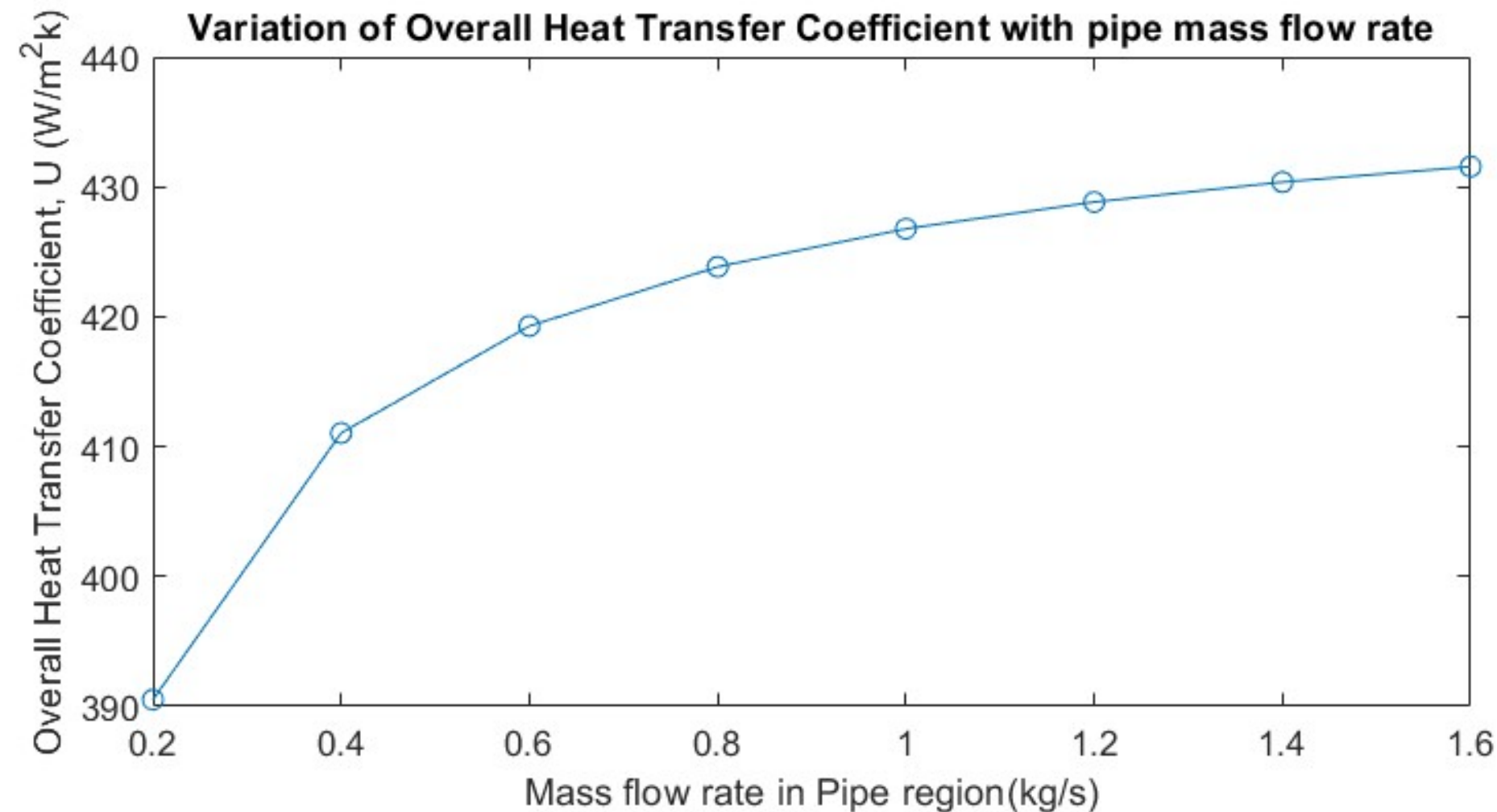
This loss is due to friction in pipe that is major loss.



Annular region pressure drop

Independent of Inner tube mass flow rate

INNER TUBE MASS FLOW RATE: OVERALL HEAT TRANSFER COEFFICIENT & TOTAL HEAT TRANSFERRED



Overall Heat Transfer

Velocity $\propto$ mass flow rate

Reynolds number, $Re \propto$ velocity

$Nu \propto Re$

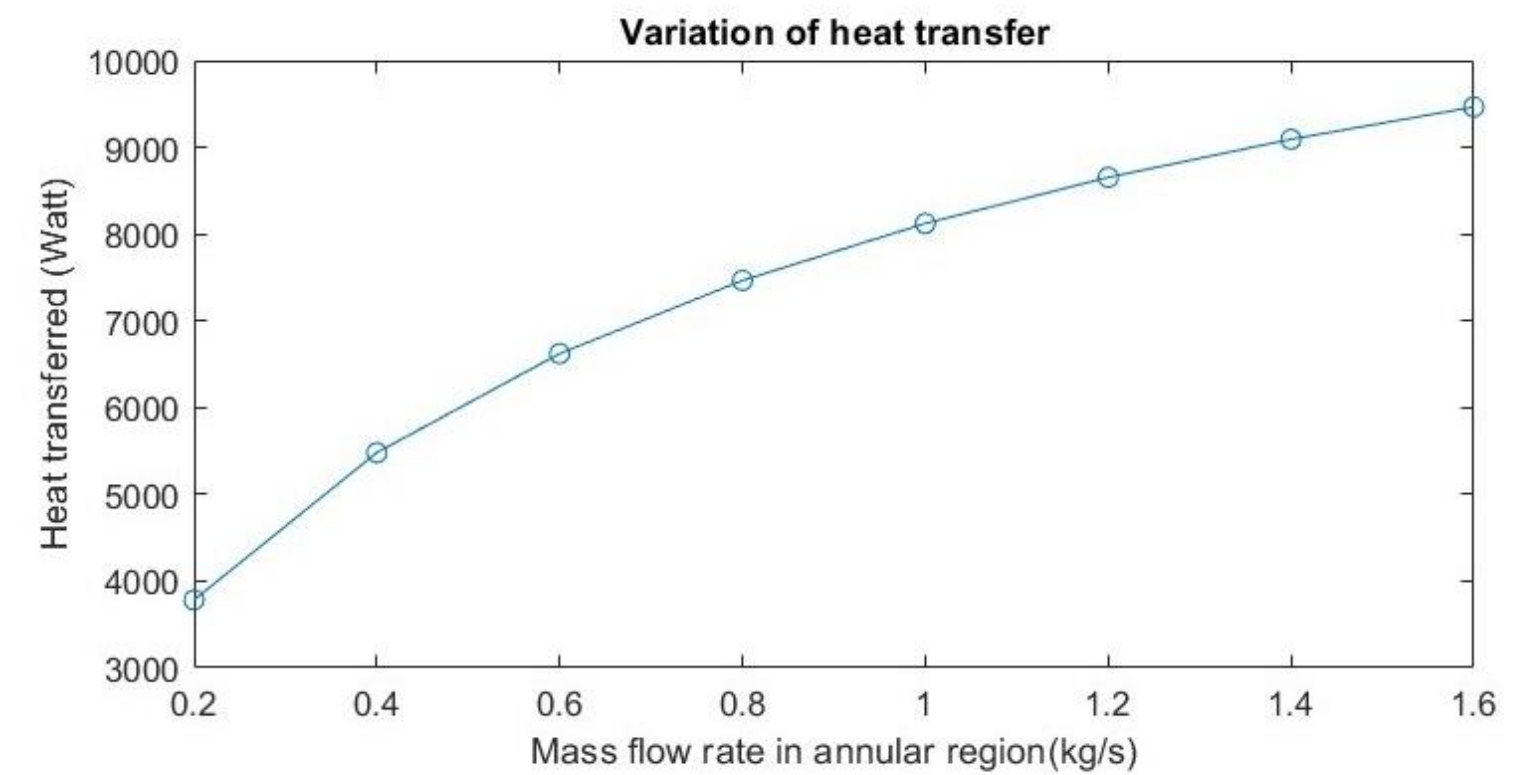
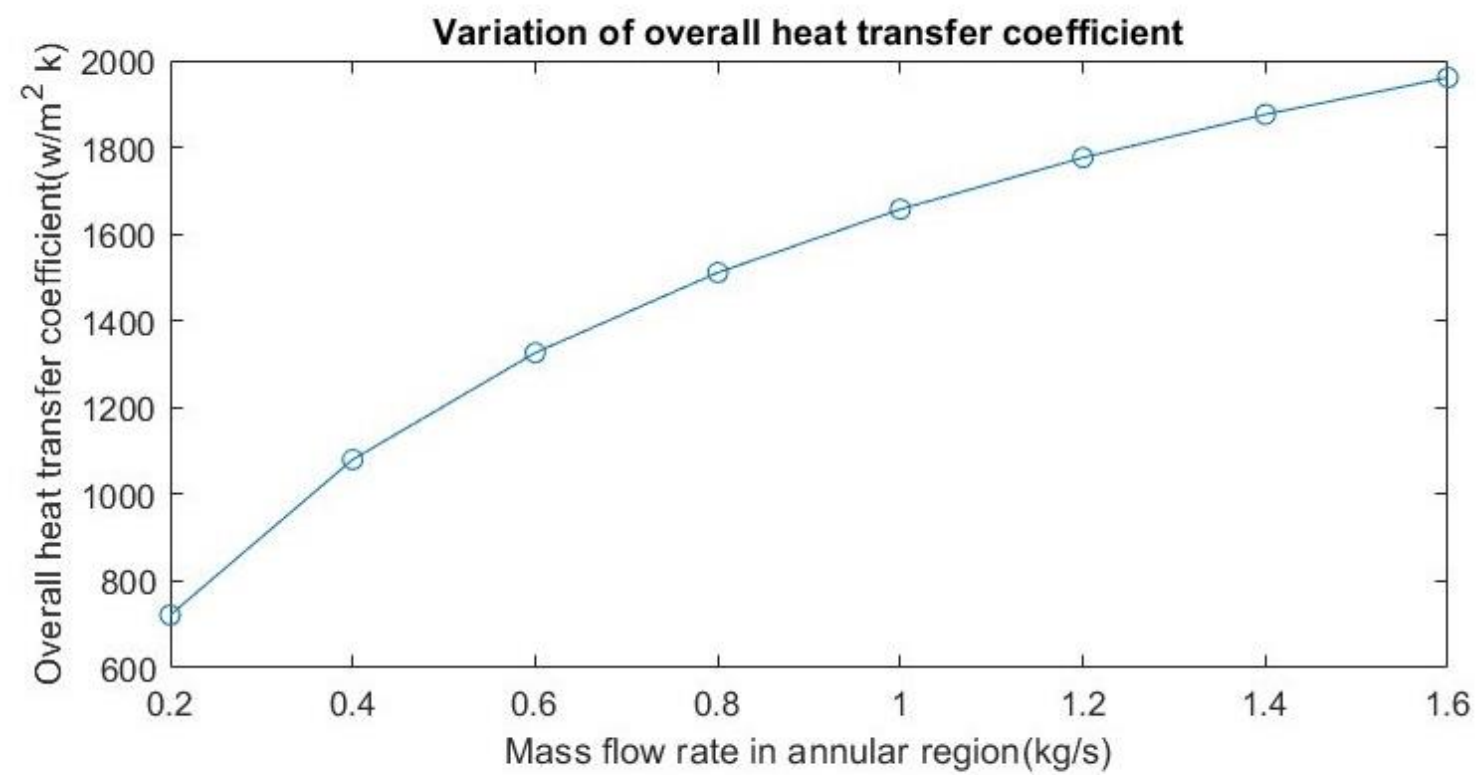
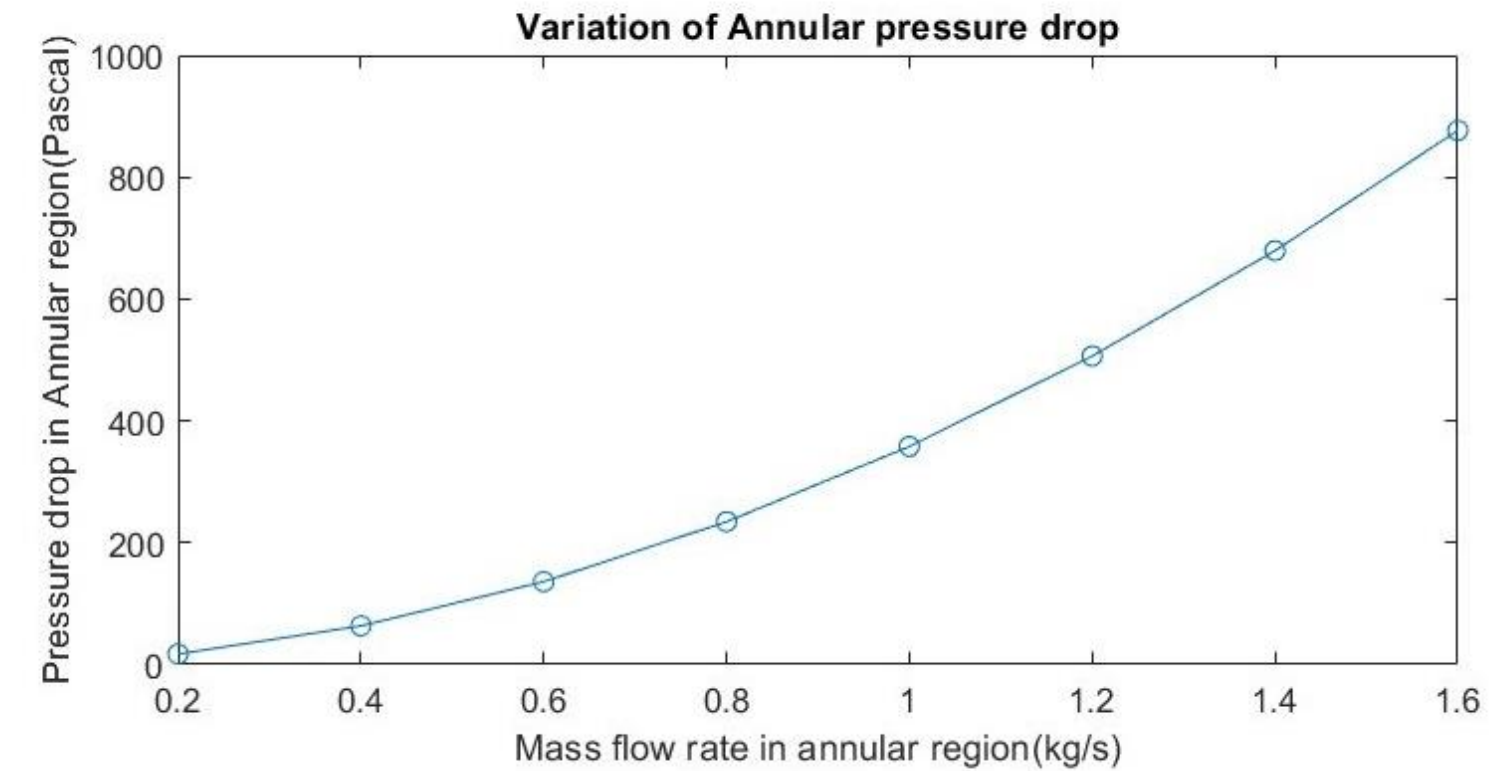
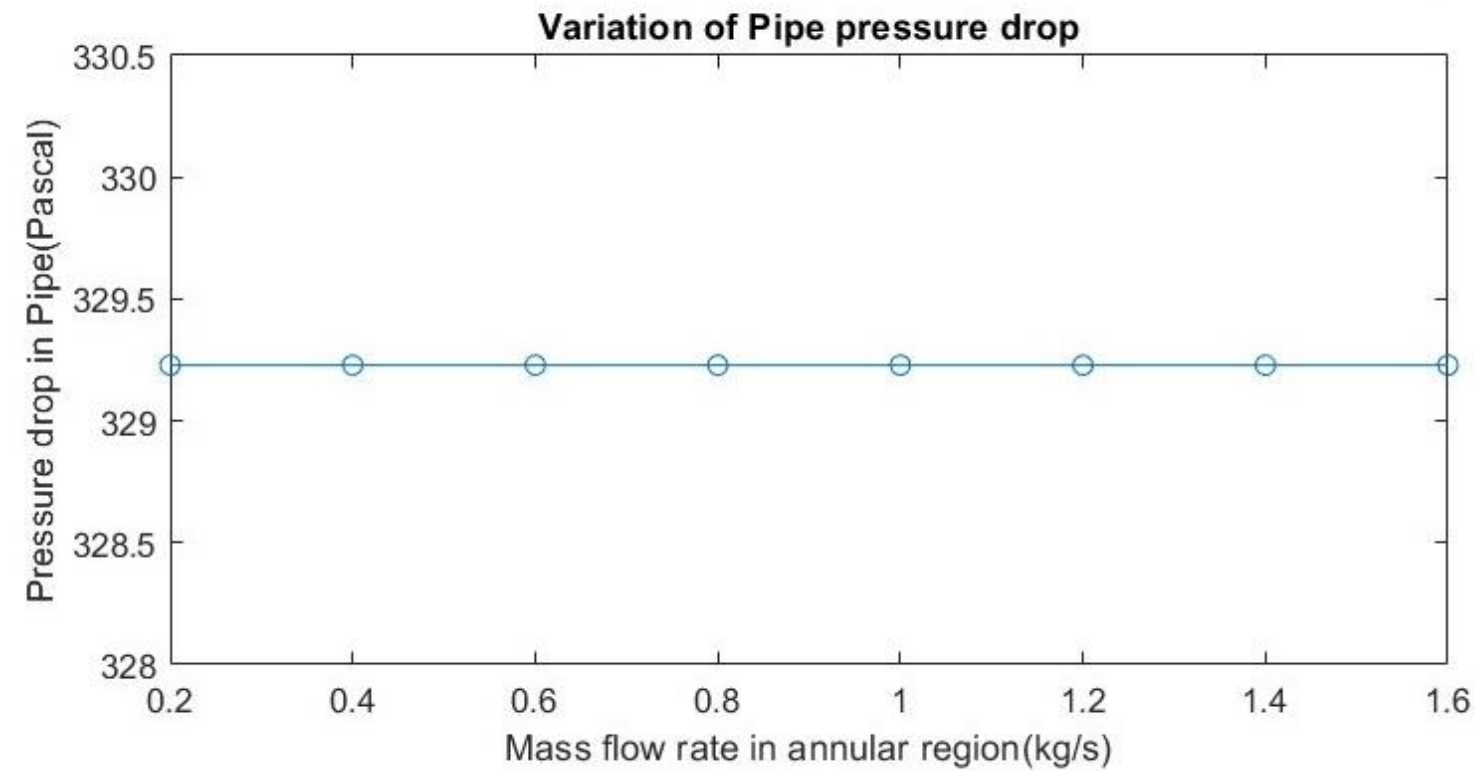
$h \propto Nu$

Total Heat Transfer

Similar reason, both have decreasing slope
so optimum value finding is required

OUTER TUBE MASS FLOW RATE

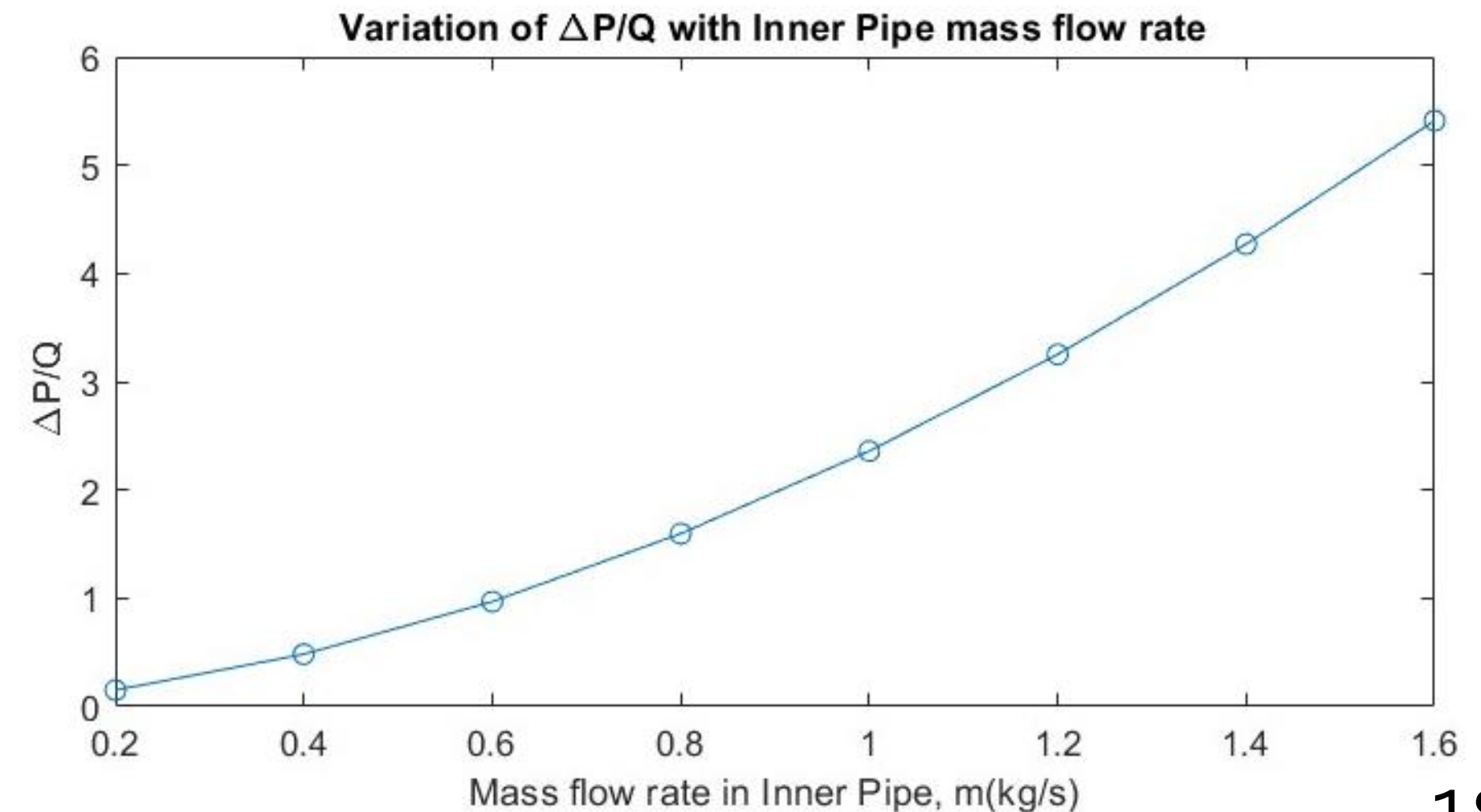
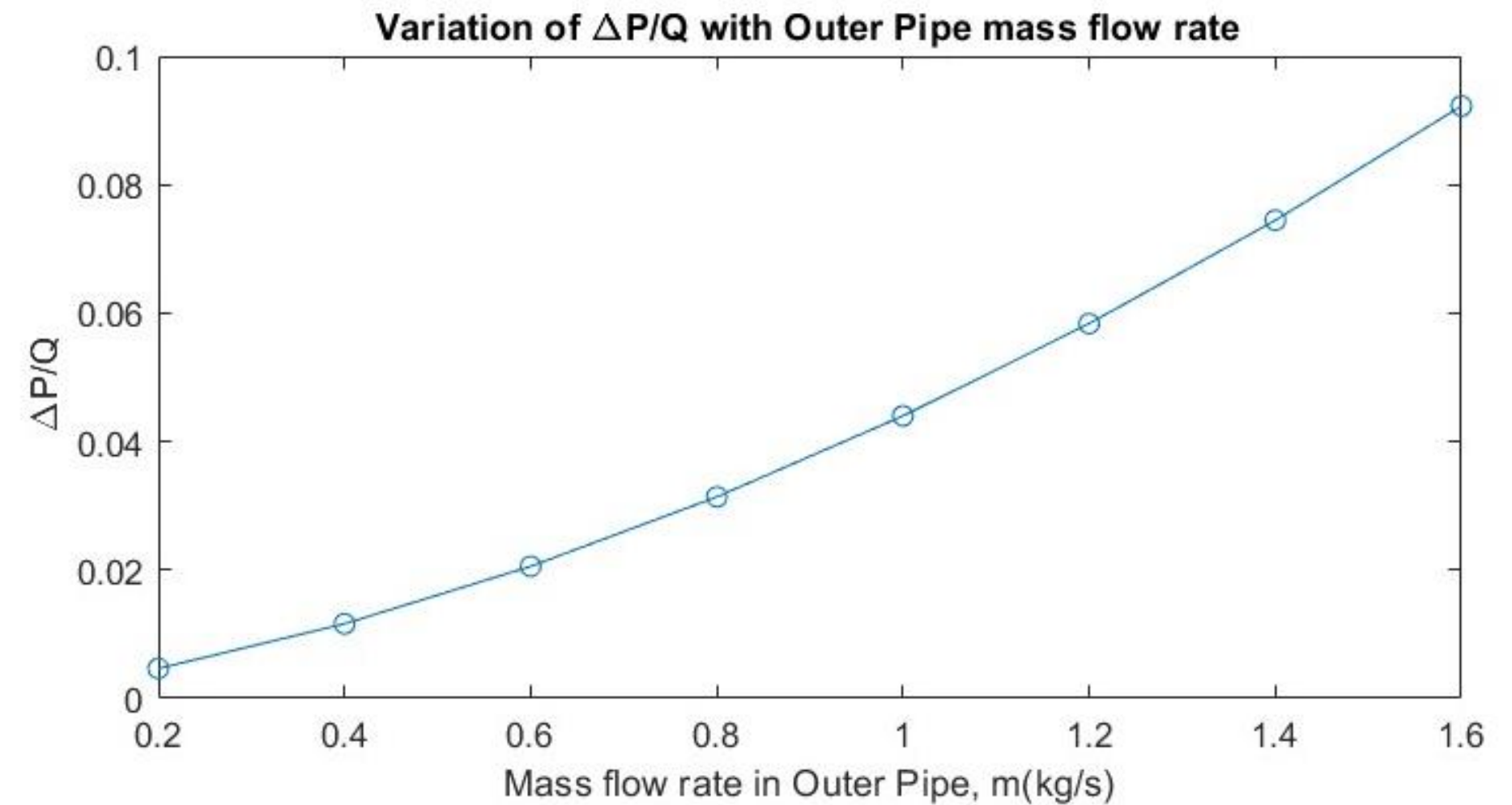
Parametric Analysis of annular Mass flow rate



SAME TREND AS BEFORE

Ratio Analysis

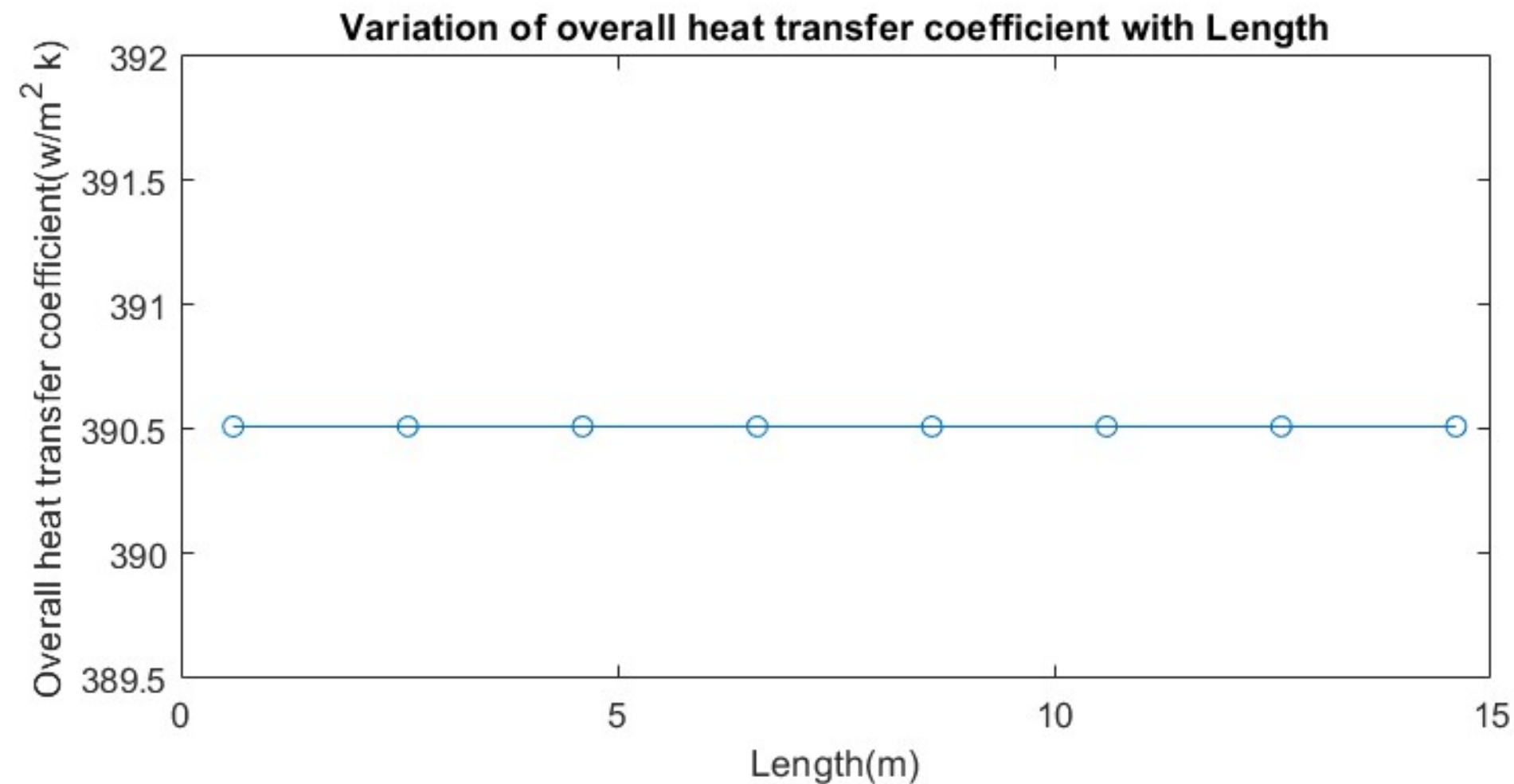
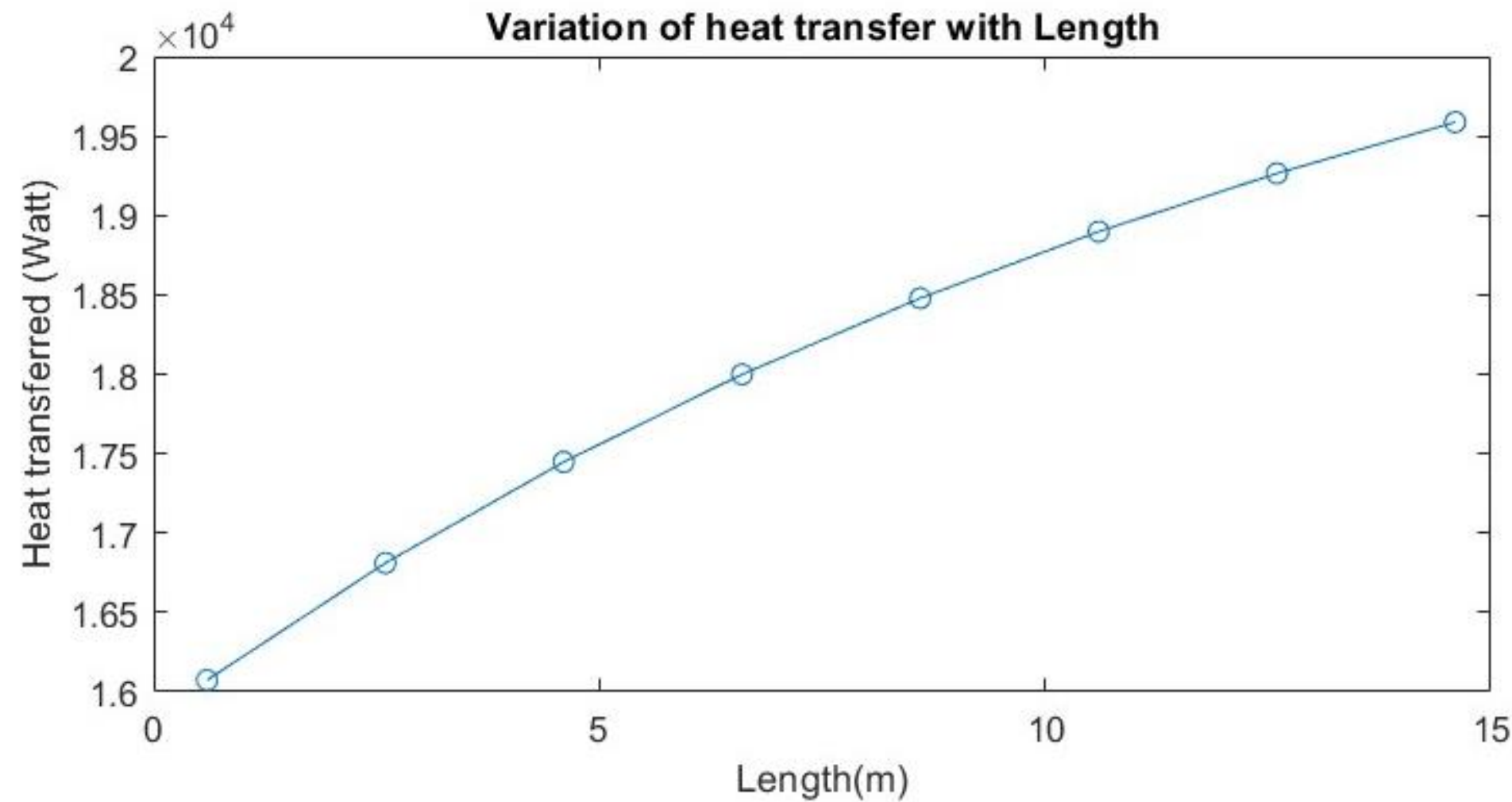
- Pressure drop & Total heat transfer both increases with increasing mass flow rate, so ratio analysis can be used as a parameter to find the optimum mass flow rate.
- The ratio is significantly higher in higher inner pipe mass flow rate than higher outer pipe mass flow rate.
- As optimum value, inner pipe mass flow rate was taken to be 0.2 kg/s
- Mass flow rate of outer region can be taken a higher value as the ratio is small comparing to inner region.



LENGTH: HEAT TRANSFERRED & OVERALL HEAT TRANSFER COEFFICIENT

$$Q \propto A \propto L$$

Positive but decreasing slope.
Optimum length must be used.



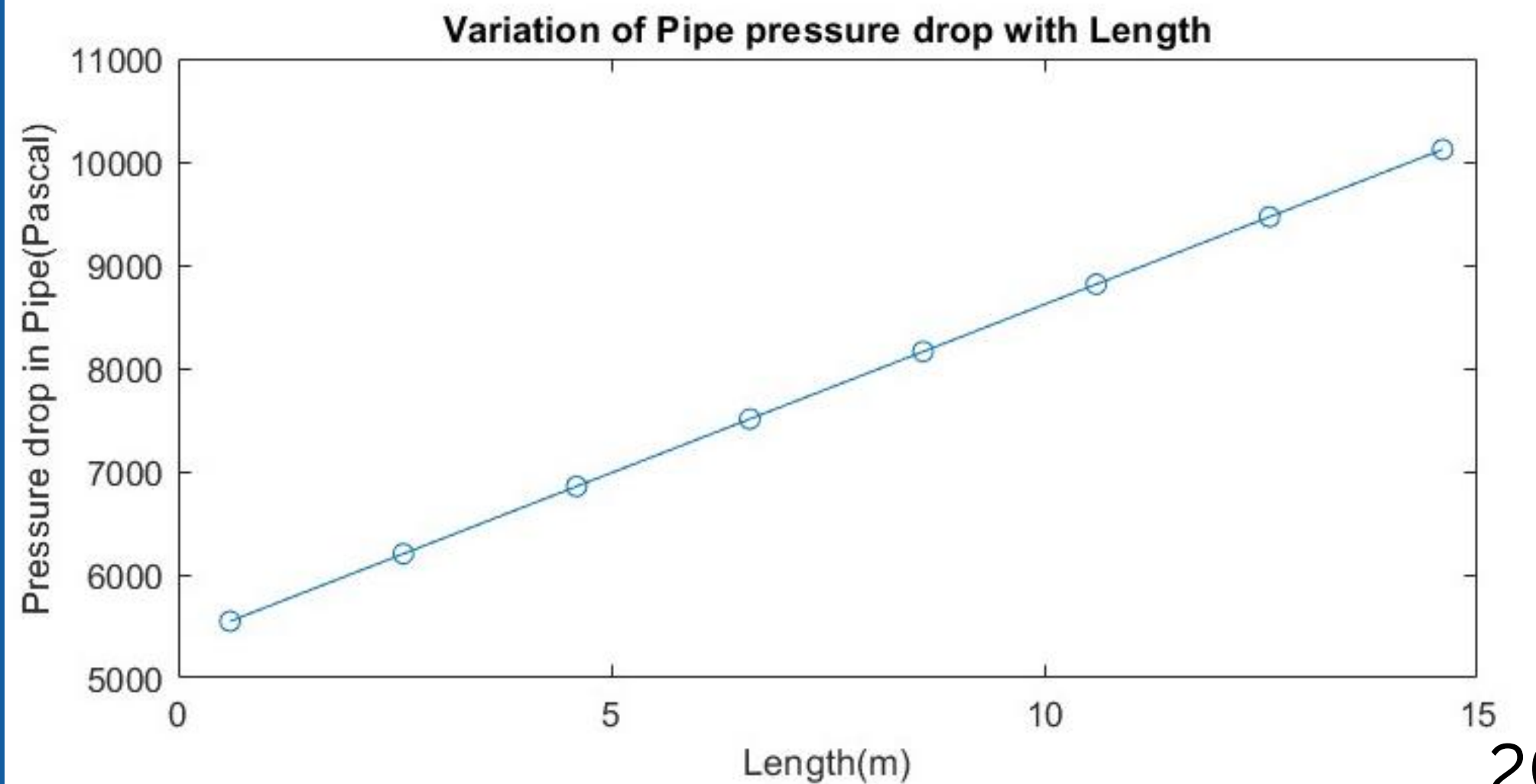
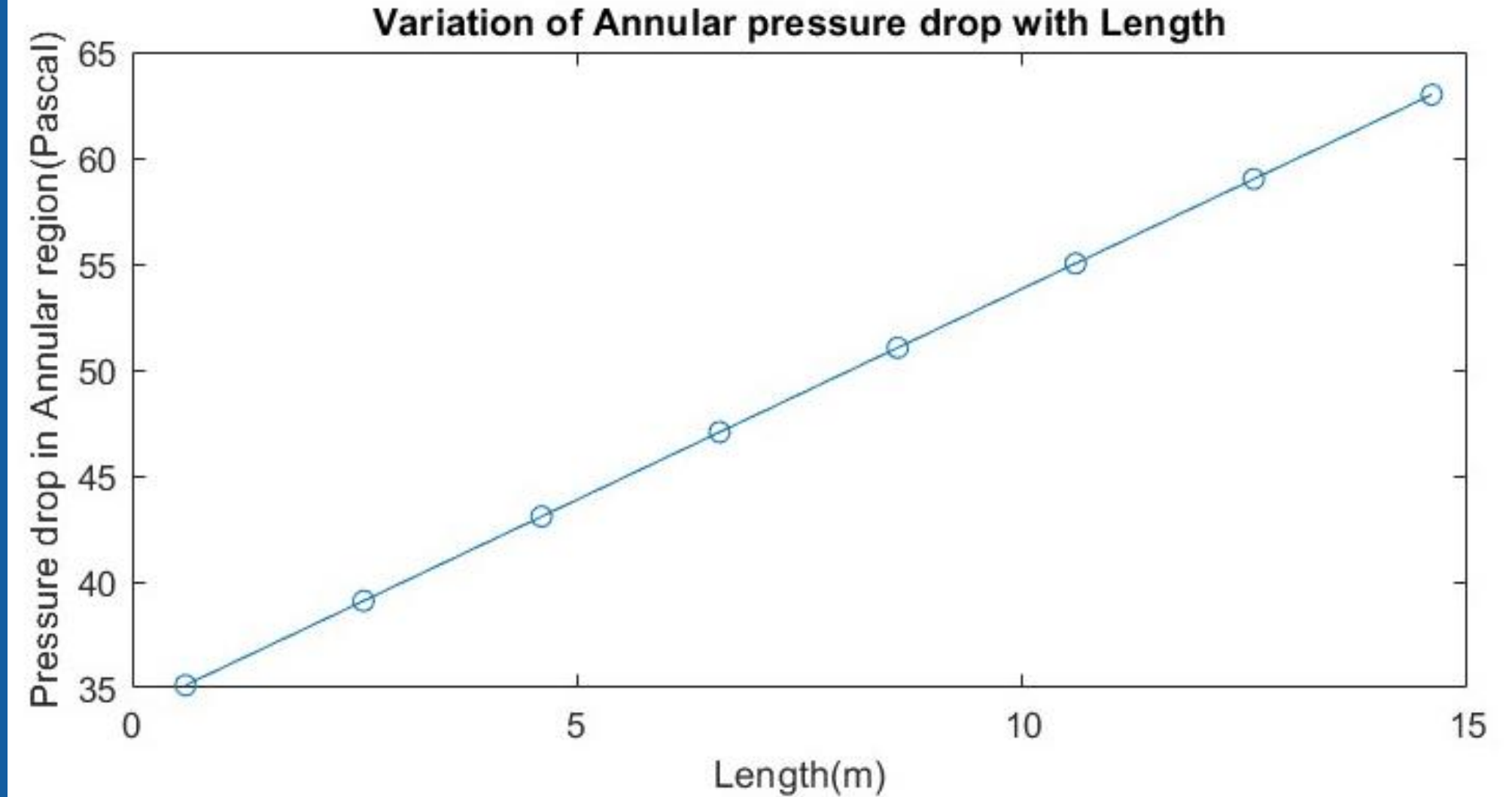
$$U \propto Nu \propto Re \propto v$$

Independent of length

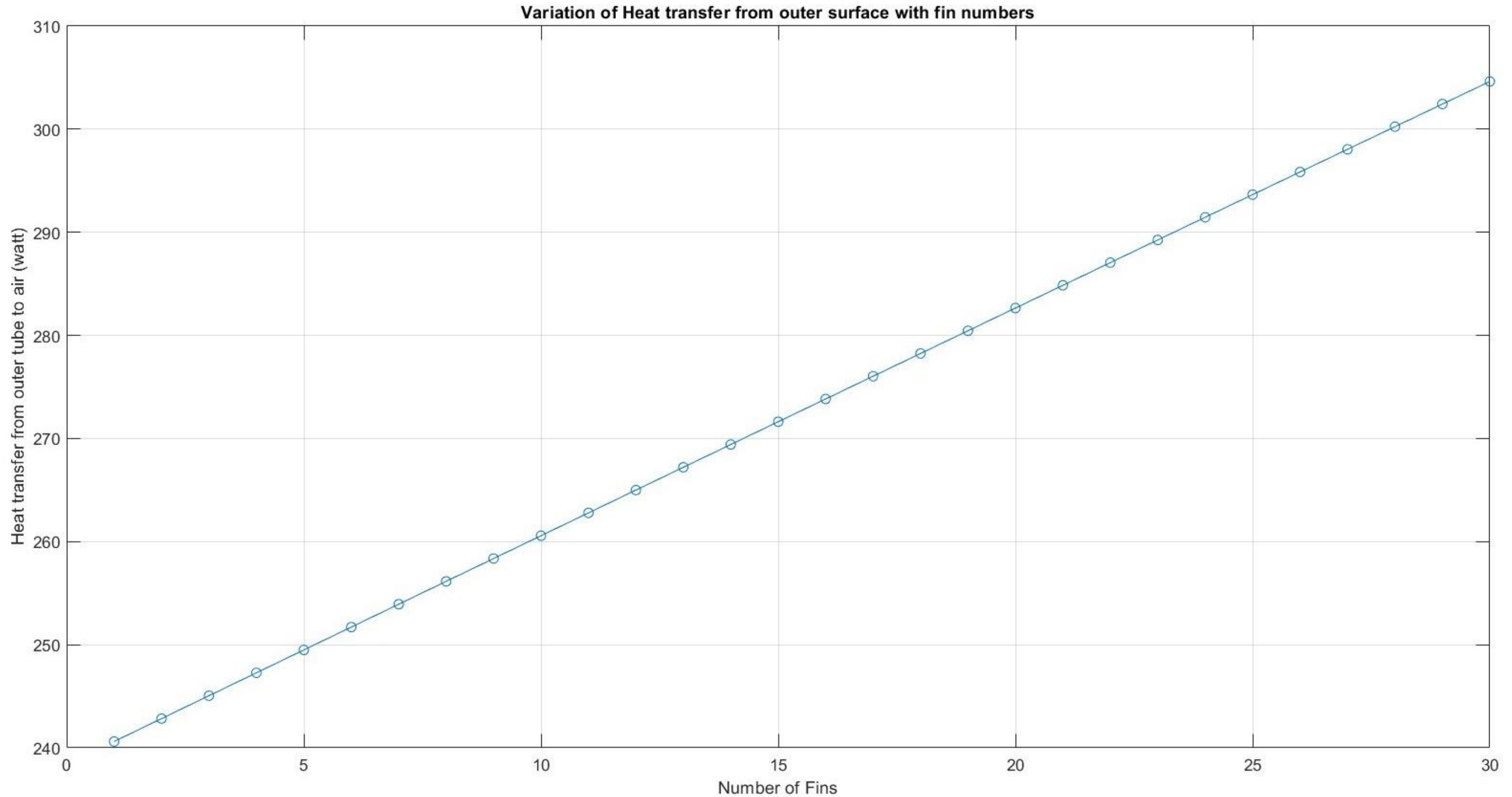
LENGTH: PRESSURE DROP

Darcy–Weisbach equation:
Pipe pressure drop $\propto$ Length

Pipe pressure drop is very high than annular
region as the flow area is reduced.



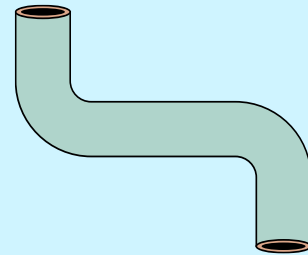
NUMBER OF FINS: HEAT TRANSFERRED



HEAT TRANSFER INCREASES AS FIN NUMBER INCREASES, WHICH WOULD BE SIGNIFICANT IF HIGH CONDUCTIVE MATERIAL WAS USED

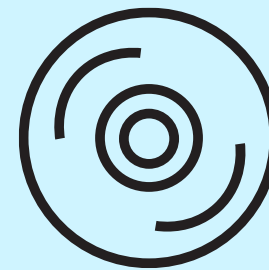
MATERIAL SELECTION

Material selection is one of the most critical decisions. Considering many factors we chose the following materials.



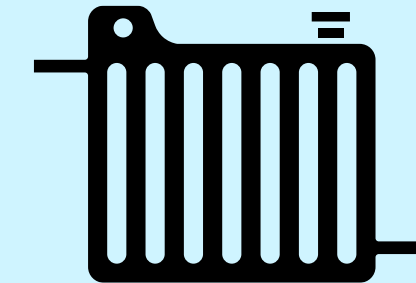
Outer and inner pipe

We have opted for mild steel for both outer and inner pipe



Turbulator

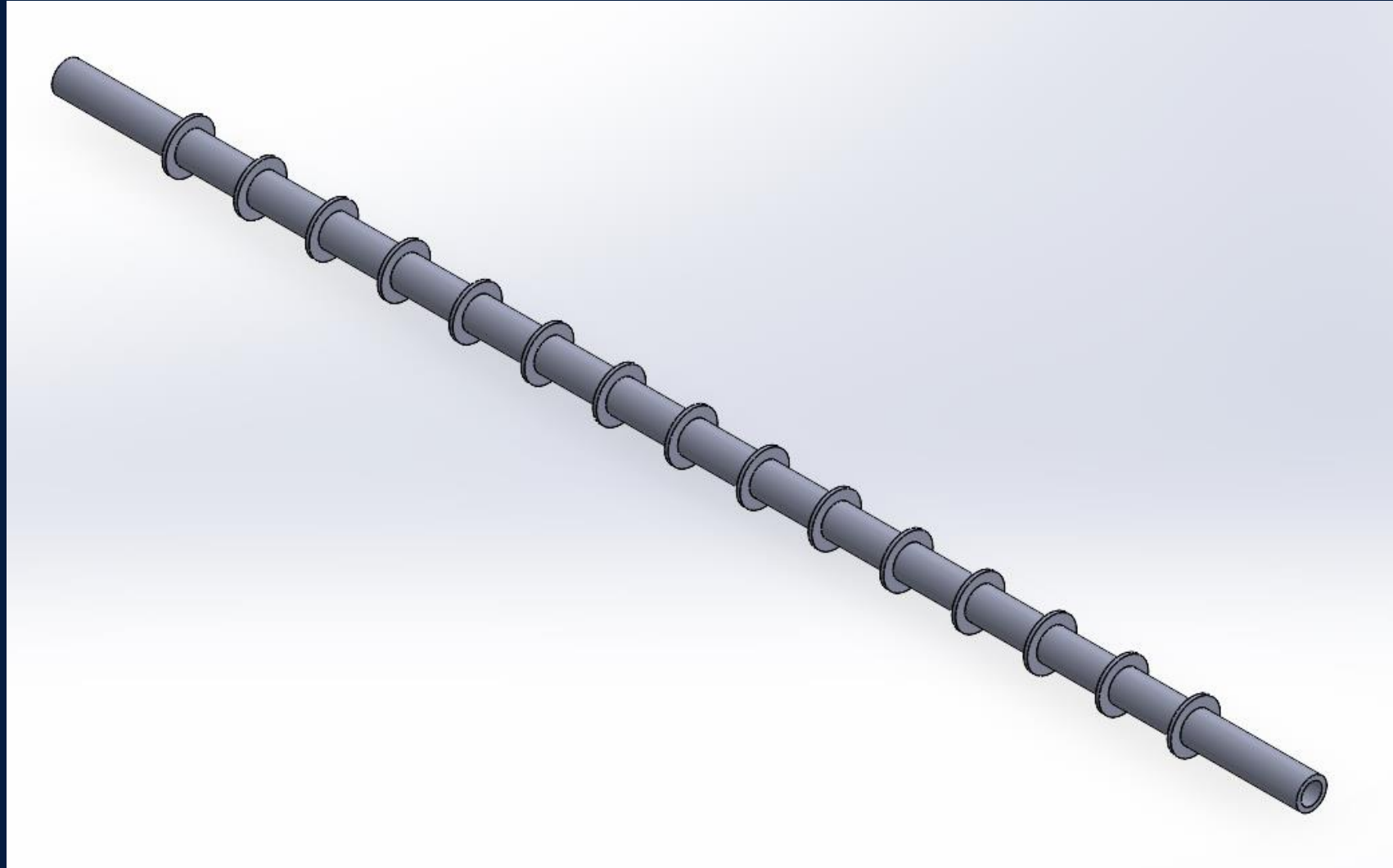
For our disk type turbulator, we chose stainless steel



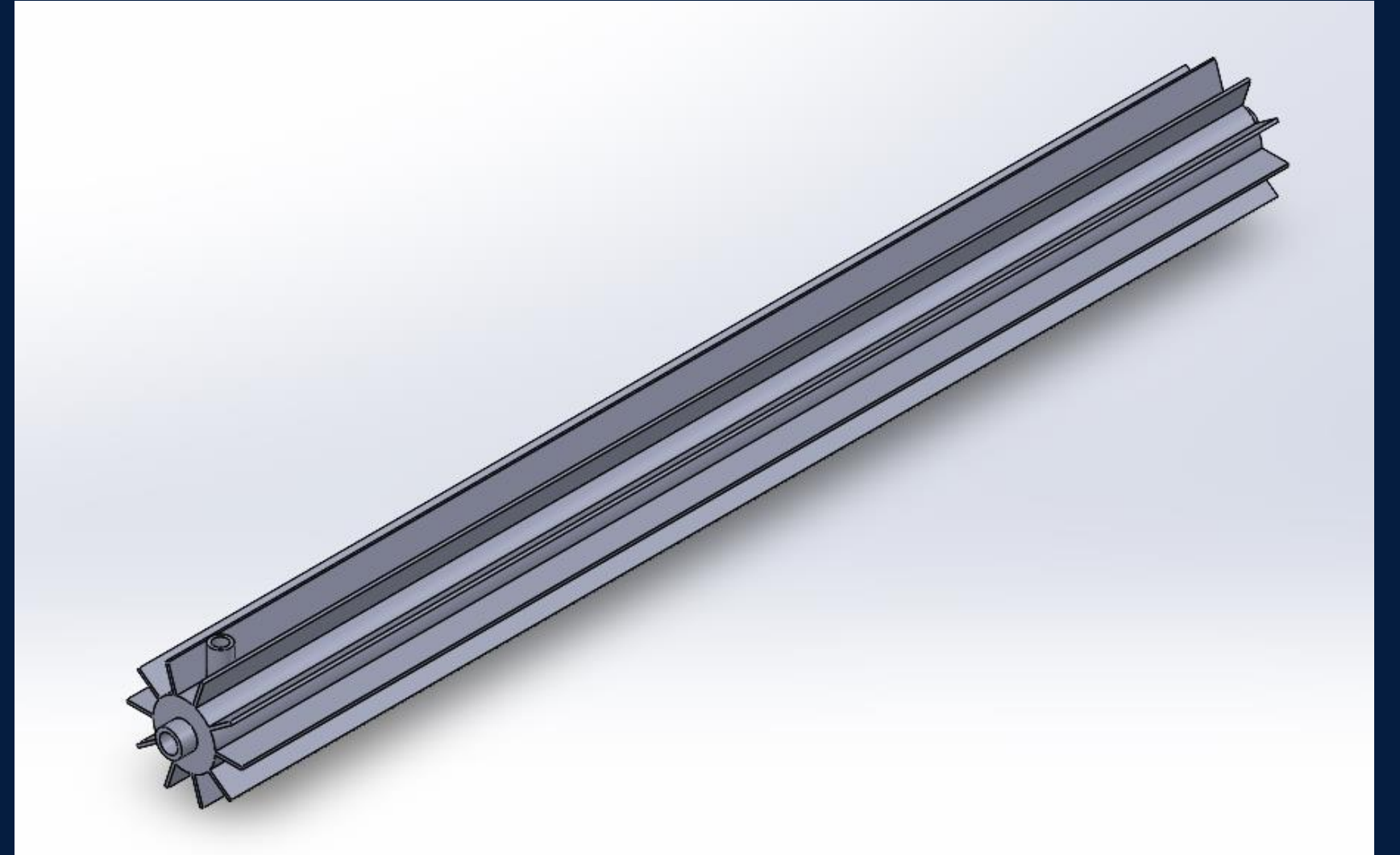
Outer Fins

We chose Mild steel (same material as pipes) for outer fins

CAD Model

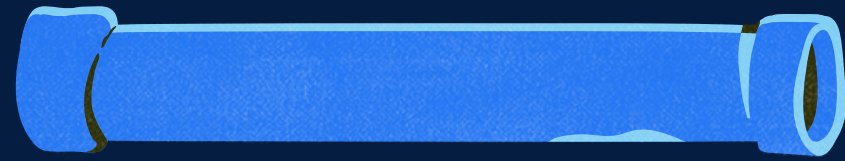


**Inner Pipe with
turbulators**



Final Assembly

MAKING OF THE PIPES



Pipe cutting

Pipes of specified length are available in market

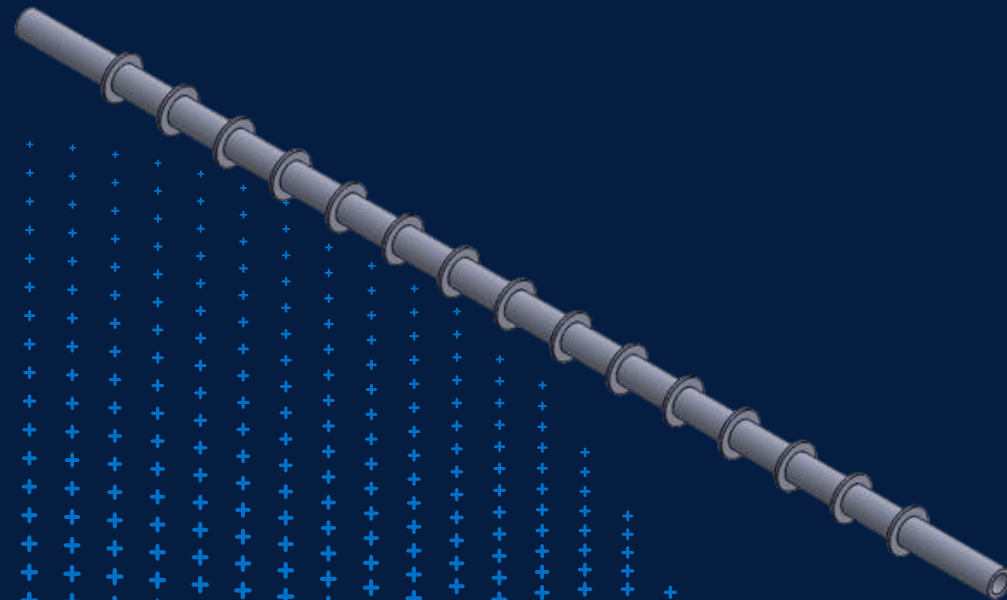
The faces are chamfered to reduce stress concentration

Face chamfering

Welding of pipes

The pipes are attached with the help of gas welding

Socket connecting



TURBULATOR FABRICATION



10 such turbulators are gas welded to the thin rod

The thin rod is then welded into the inner pipe

Steel disk choosing

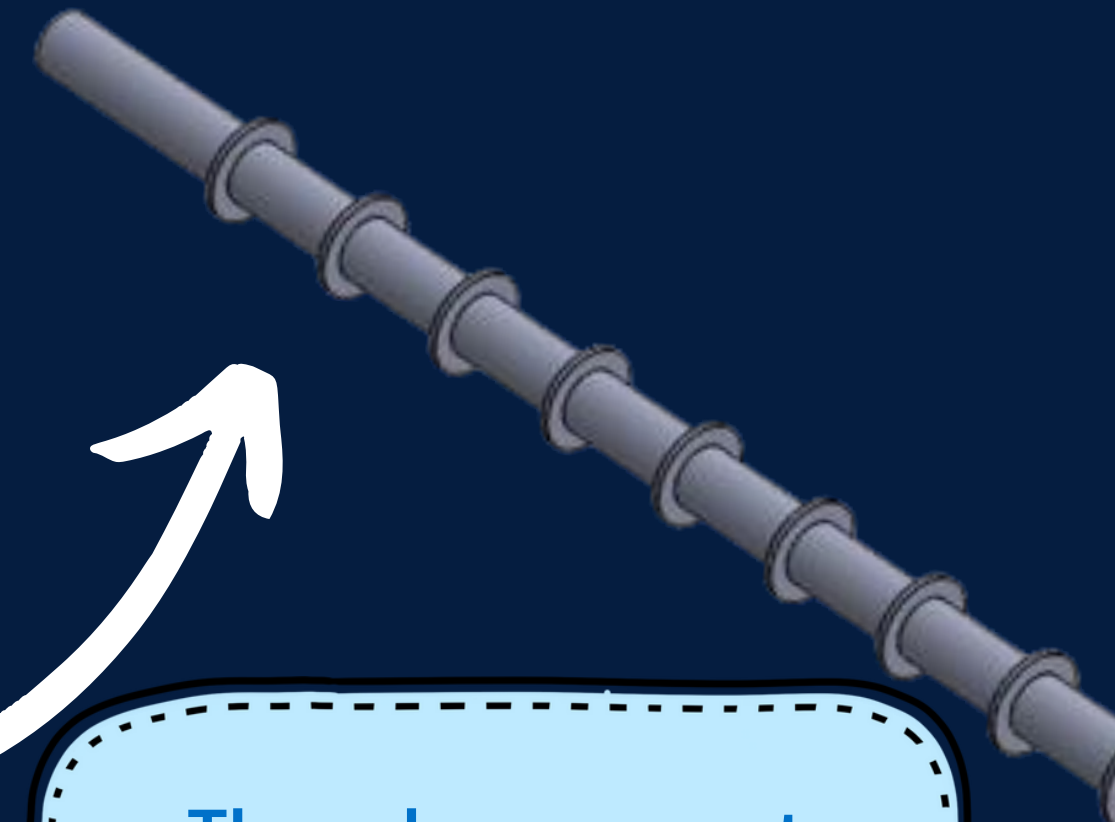
Drilling the center

Cutting the edges

Welding the centers to a rod

Inserting the rod into the inner piper

The inner hole in the turbulator is made by drilling



The edges are cut with the help of a hand cutting tool

ATTACHING THE FINS

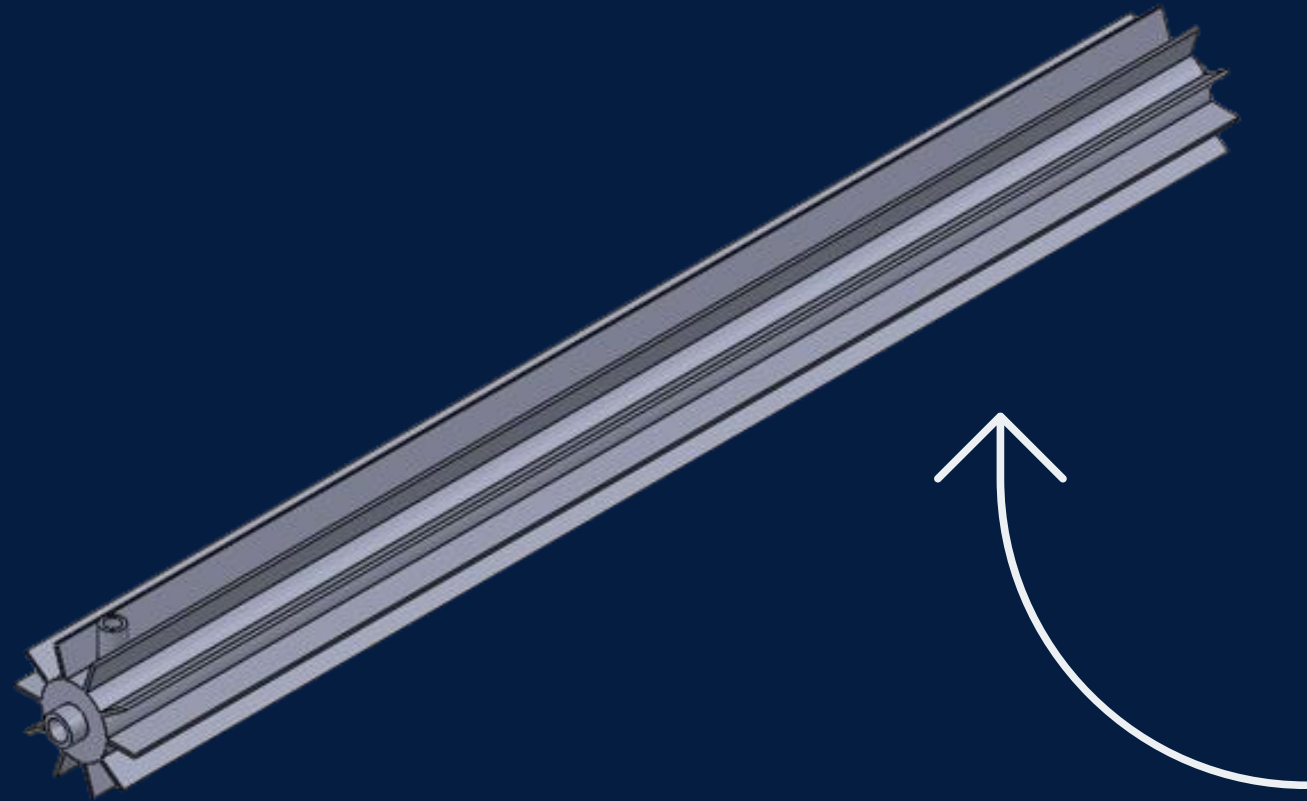


Plate of steel
cutting

8 similar
rectangular fins

Arc welding of
the fins

Effective for forced
convection

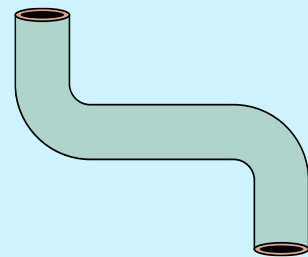
The fins are welded
to outer pipes unlike
traditional fins

The material same as
the pipes are used to
make the fins

The outer fins only
work when forced
convection occurs

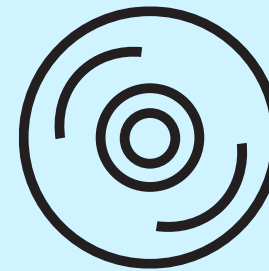
LIMITATIONS

Our projects has many limitations due to costing, availability of material.



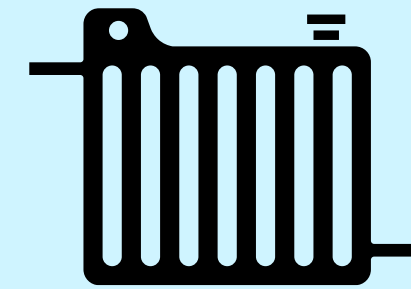
Outer and inner pipe

No proper holding fixture.
Cant be disassembled for
maintenance and repair



Turbulator

The turbulator might
obstruct the flow in
case of high mass flow
rate and increase
pressure drop



Outer Fins

- 1.No metal casting
2. Contact resistance
3. Discountinous weld
4. Forced convection

POSSIBLE SOLUTIONS

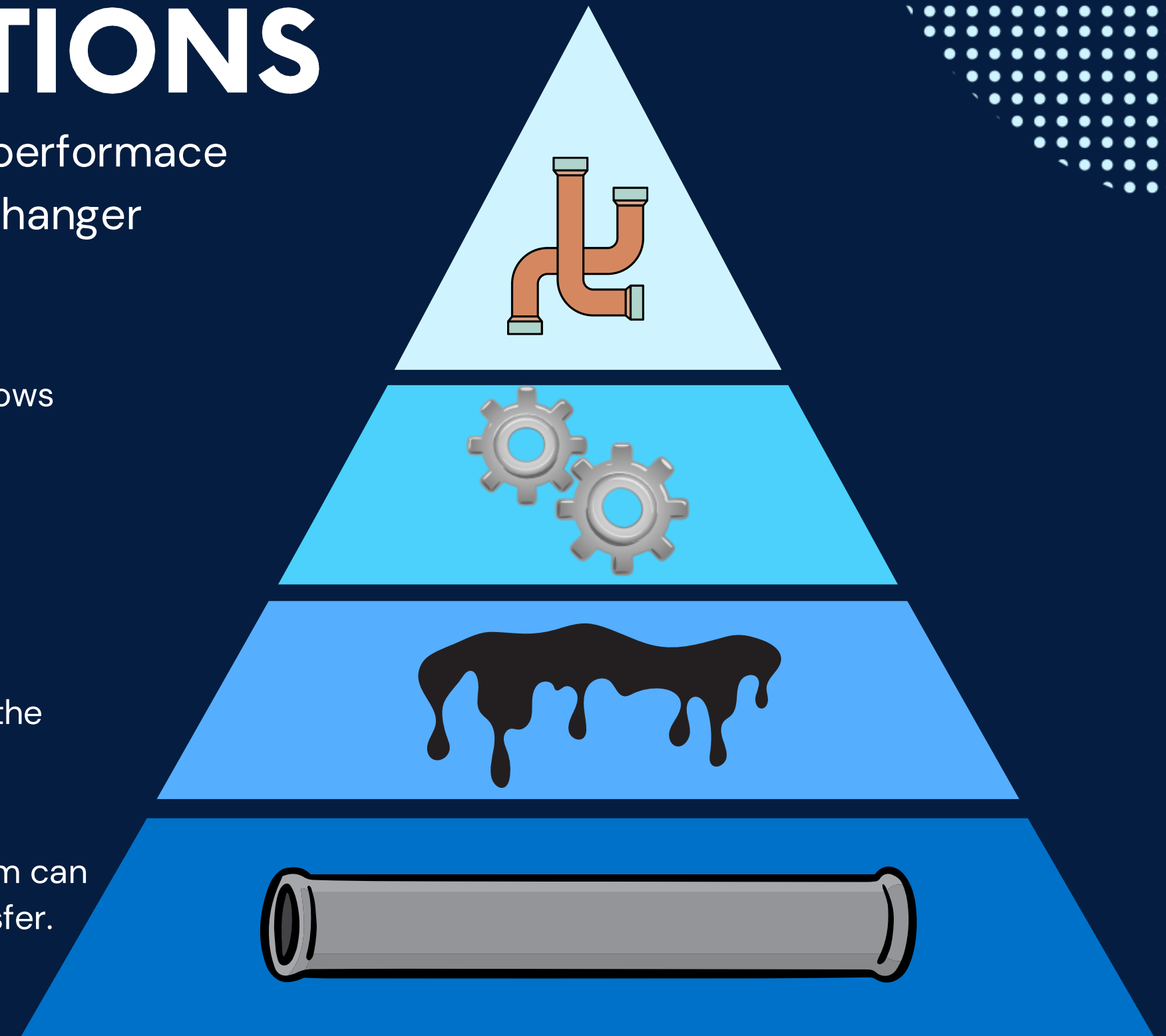
These mitigations strategies enhance the performance and solve the issues regarding the heat exchanger

01 An alternate type turbulator can be used that allows more space for fluid to flow.

02 The fins can be connected by a continuous welds running along the entire length

03 If the fins are made from metal casting process, the limitations due to welding would be removed.

04 More conductive metal like copper and aluminium can be used in the pipes order to enhance heat transfer.



NUMERICAL ANALYSIS

Boundary conditions

Uniform inlet velocity profile

Inlets are assumed as
velocity inlet

Outlets are assumed as
pressure outlet

The fluid properties are
uniform

Outer wall was assumed as
adiabatic

NUMERICAL ANALYSIS

Assumptions

Parameters	Inner Pipe	Outer pipe
Inlet Temperature	25 C	90 C
Mass flow rate	0.08 Kg/s	0.2 Kg/s
Outlet pressure	0 Pa (gauge)	0 Pa (gauge)

NUMERICAL ANALYSIS

Geometry

Parameters	Inner Pipe	Outer pipe
Length	1 m	1 m
OD	1 in	2.375 in
Wall thickness	3 mm	4 mm

NUMERICAL ANALYSIS

Meshing

01

Sizing

The size of the mesh was **no more than 5mm**

02

Method

Hexahedral mesh was generated

03

Mesh Refinement

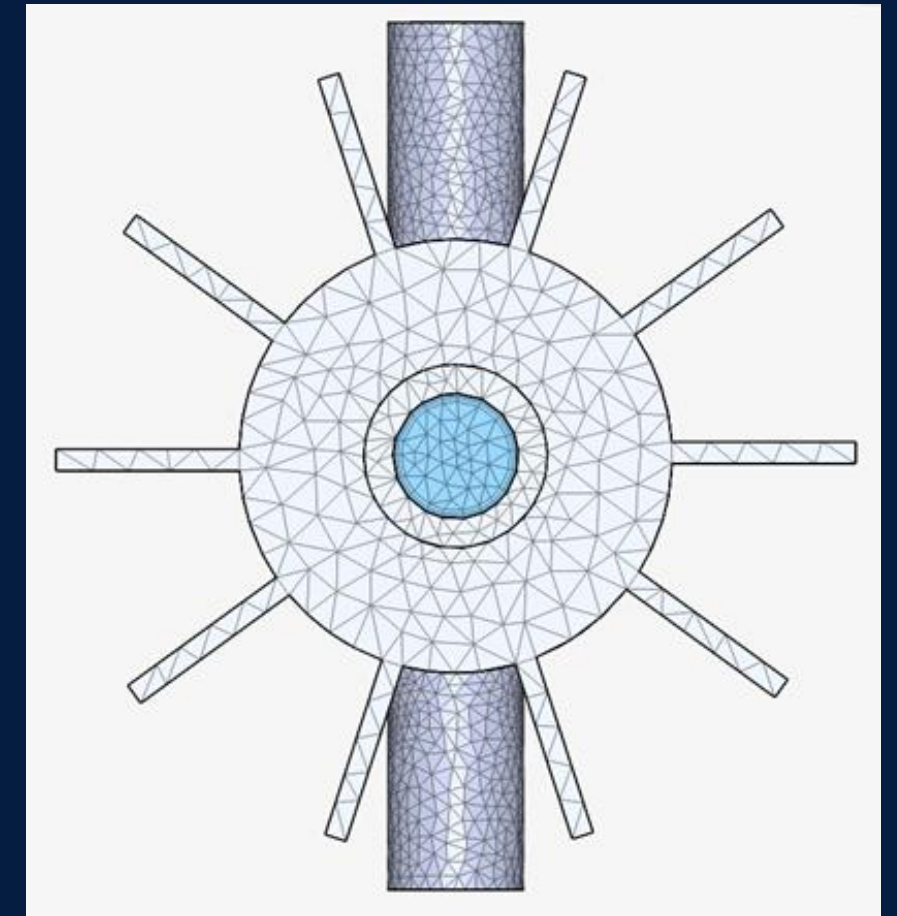
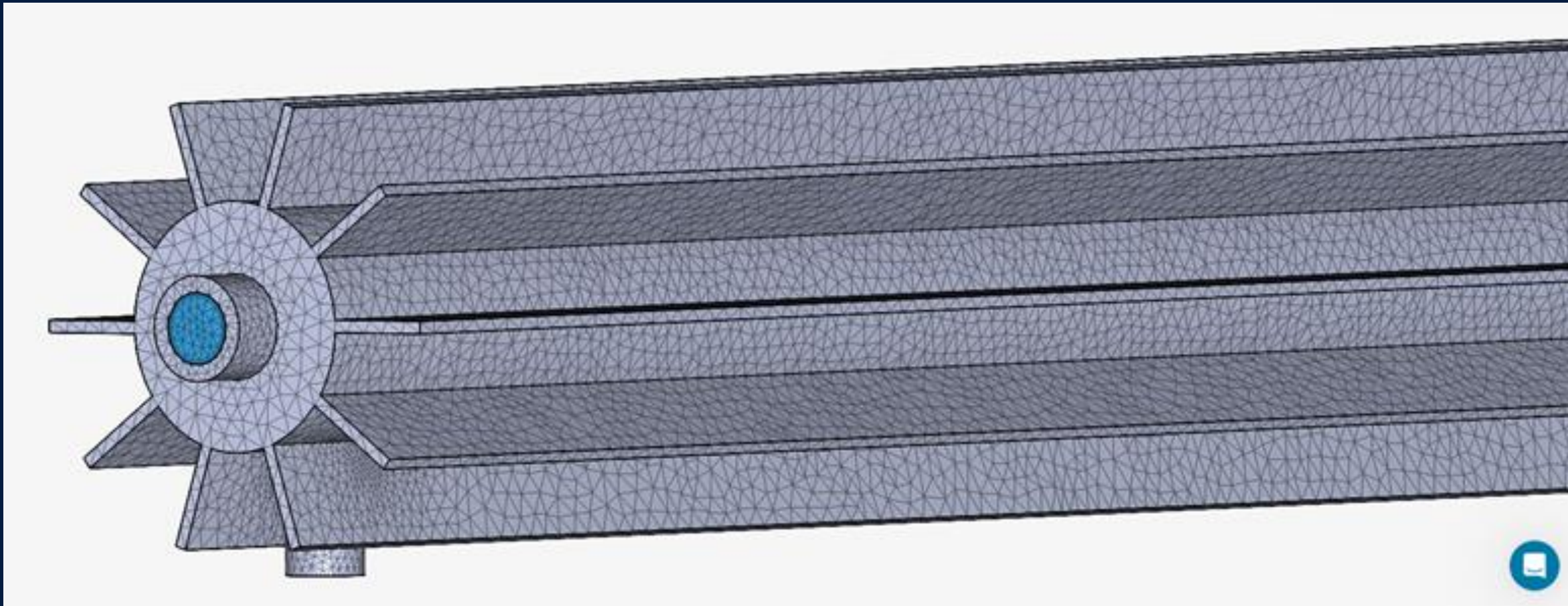
Inflation layer on the walls was used to capture the boundary layer effect

04

Statistics

The number of cells is **727.7k** and the number of nodes is **205k**

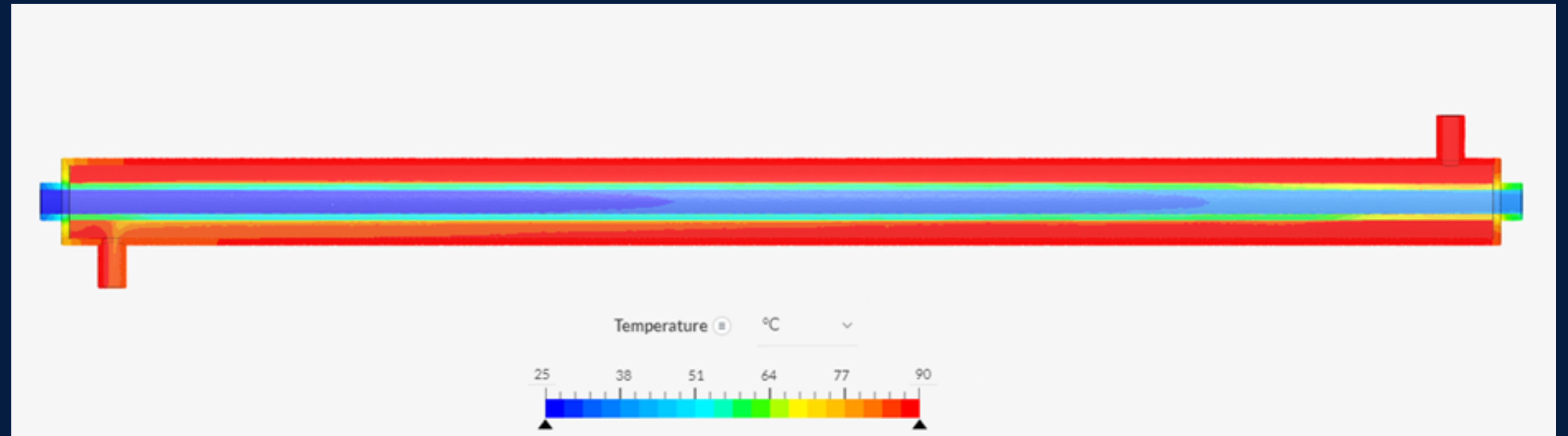
Meshing



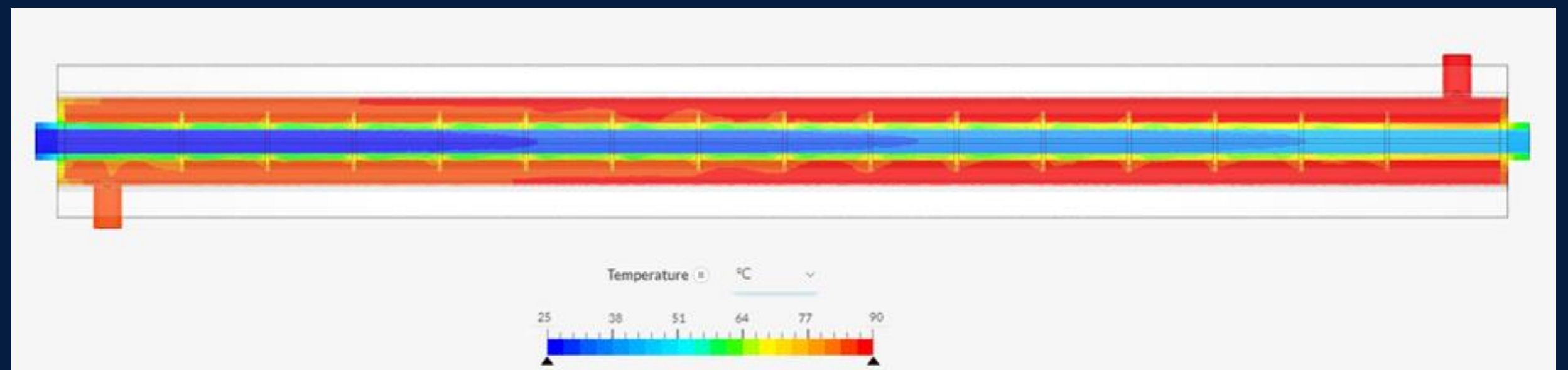
Result

Temperature

Without
Turbulator

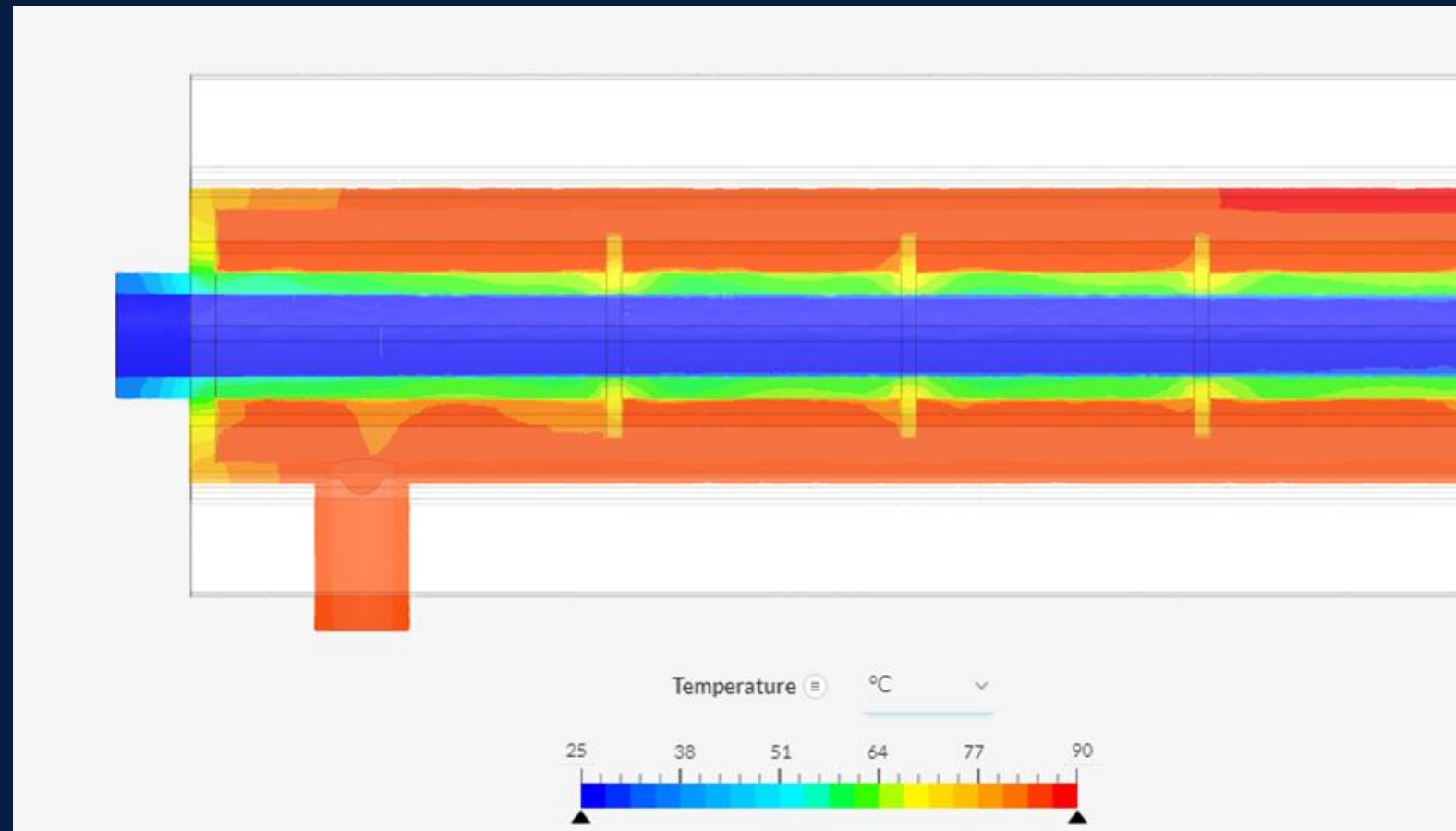


With
Turbulator

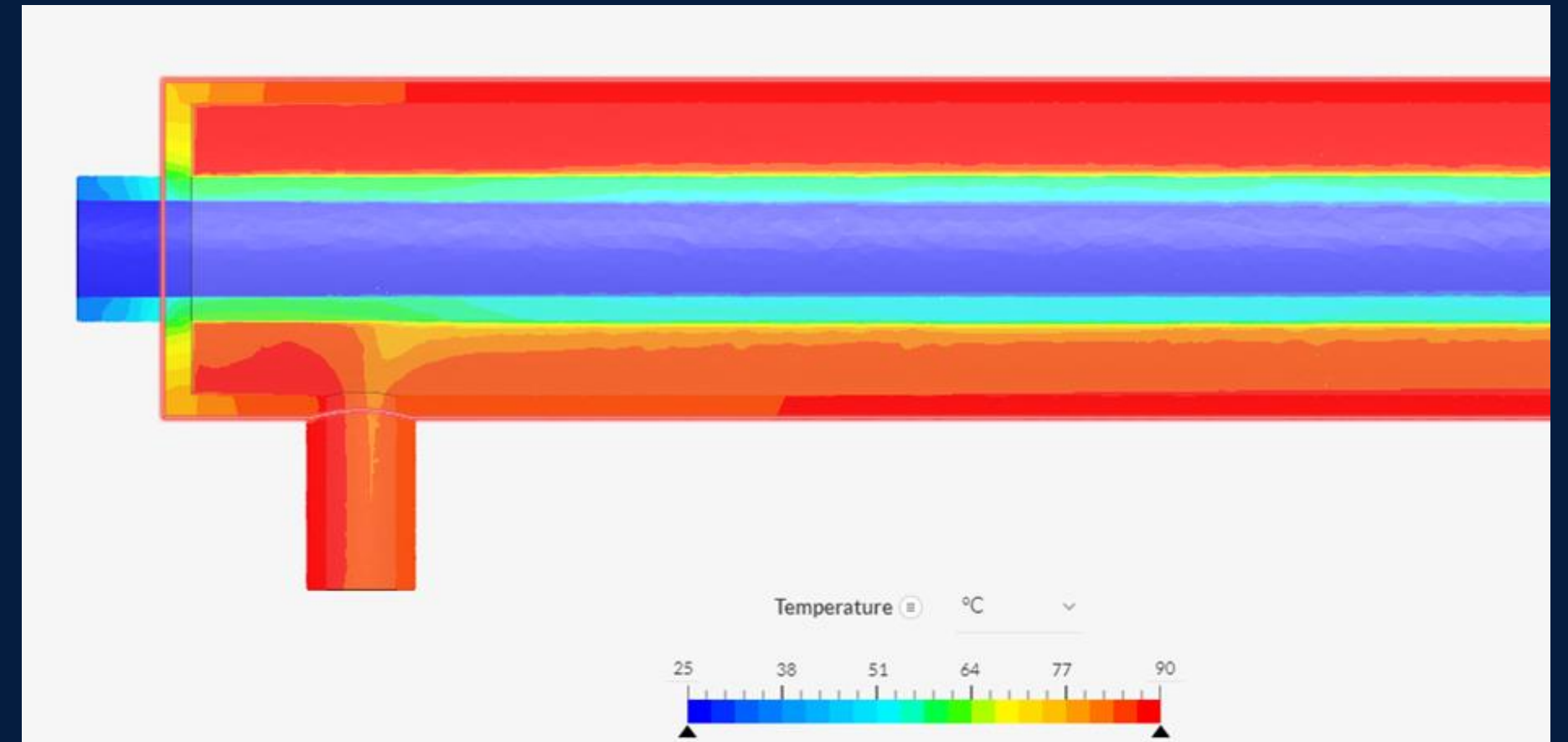


Result

Temperature



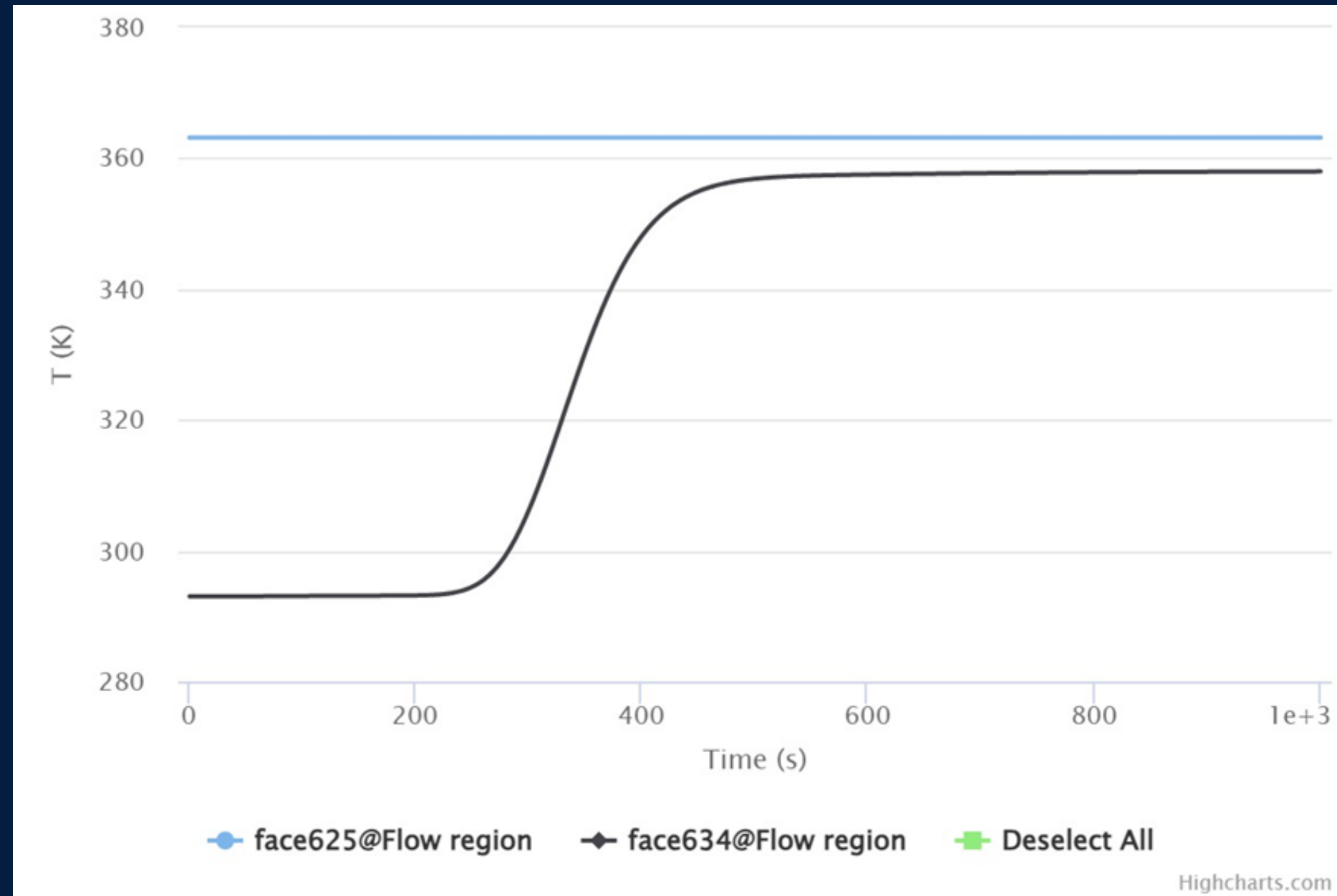
**With
Turbulator**



**Without
Turbulator**

Result

Outlet Temperature

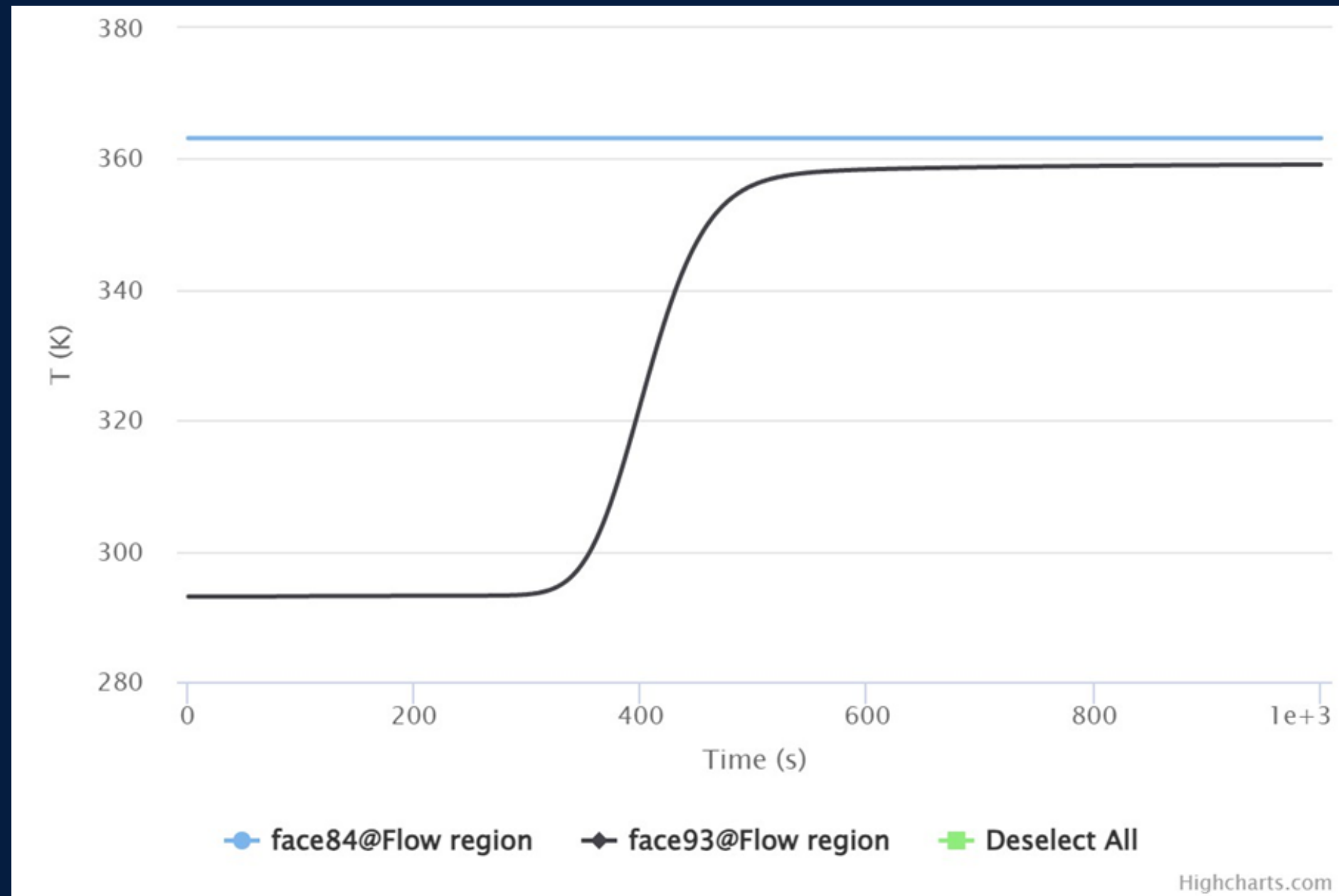


**Without
Turbulator**

Hot fluid outlet
temperature
approximately 86°C .

Result

Outlet Temperature

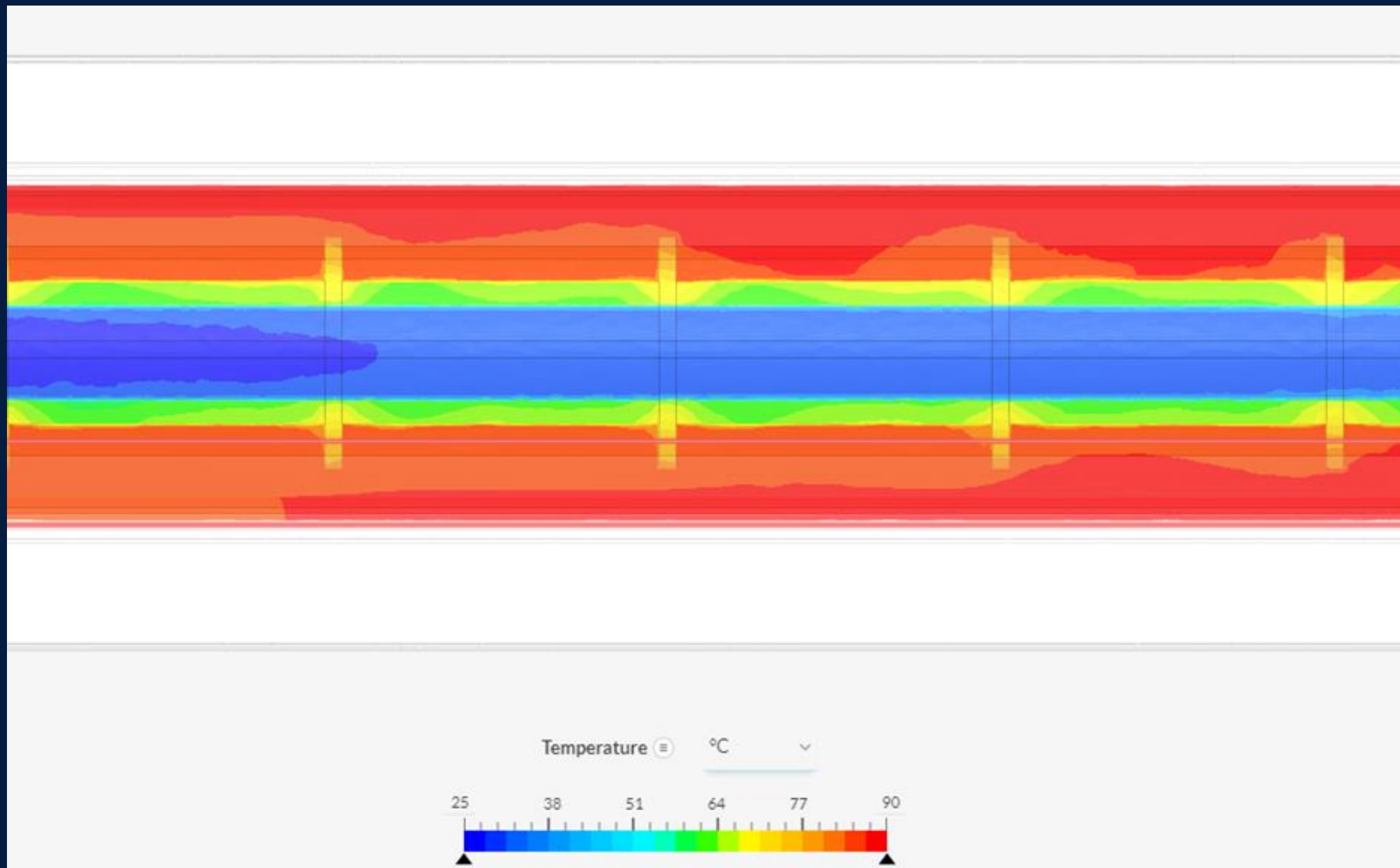


With Turbulator

Hot fluid outlet
temperature
approximately 84°C .

Result

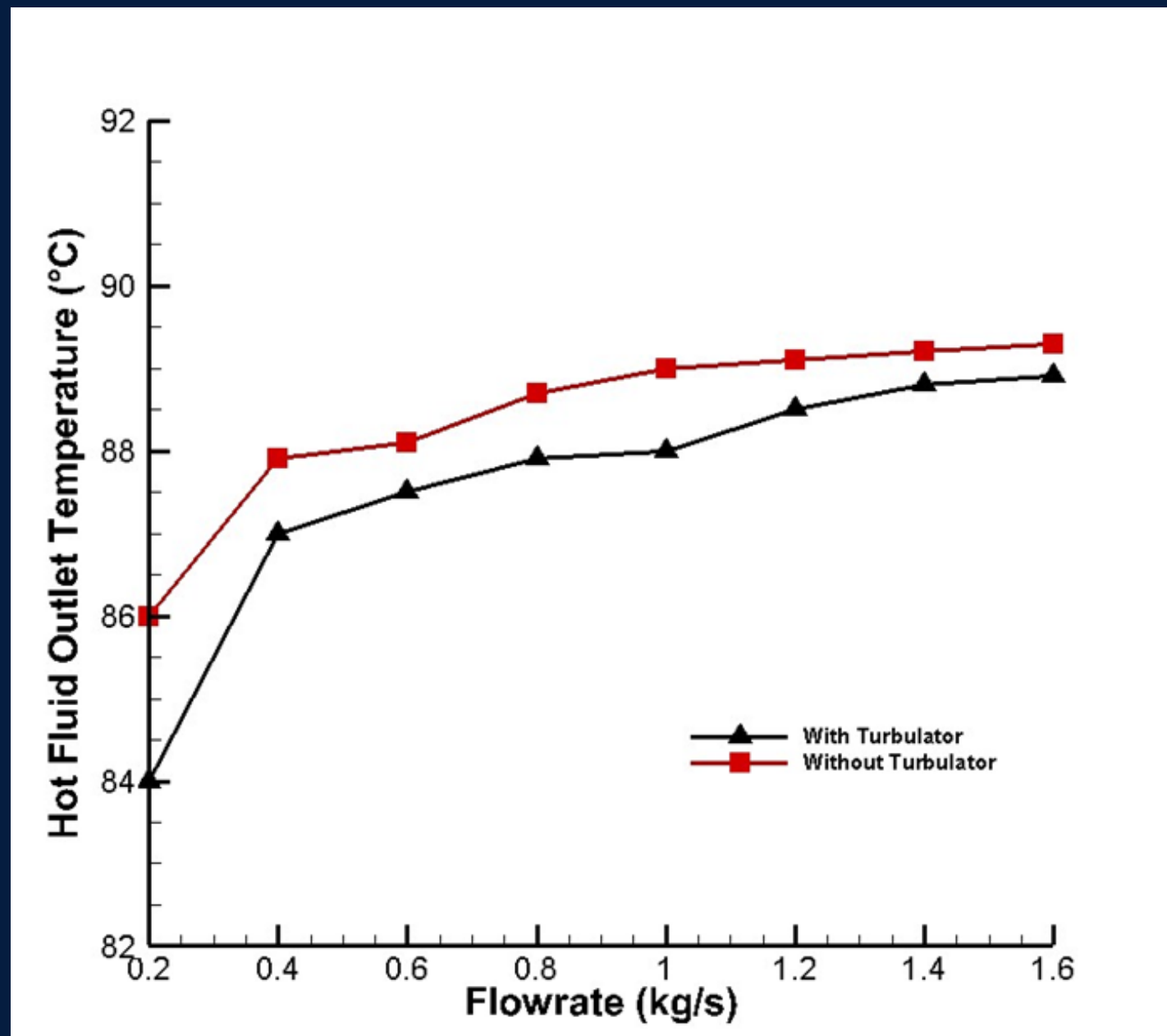
Effect of turbulator



- Creates turbulence in water flow
- Helping to mix water properly
- More Heat transfer
- Better performance

Parametric Analysis using CDF

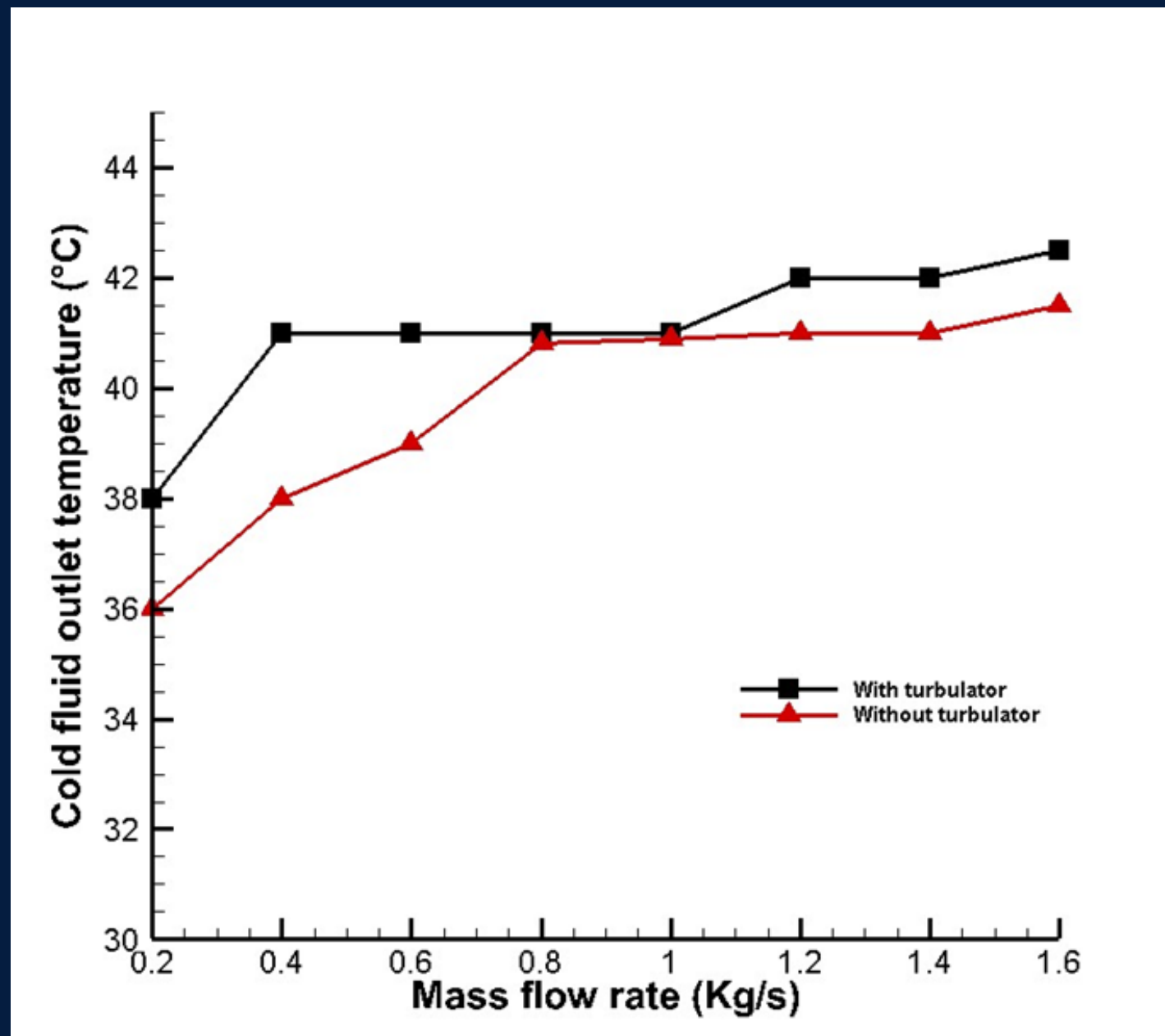
Varying mass flow rate



- The mass flow rate in the annular area (hot fluid) was changed keeping the flow rate of the internal area fixed (0.08 Kg/s).
- The hot fluid outlet temperature increases with the mass flow rate

Parametric Analysis using CDF

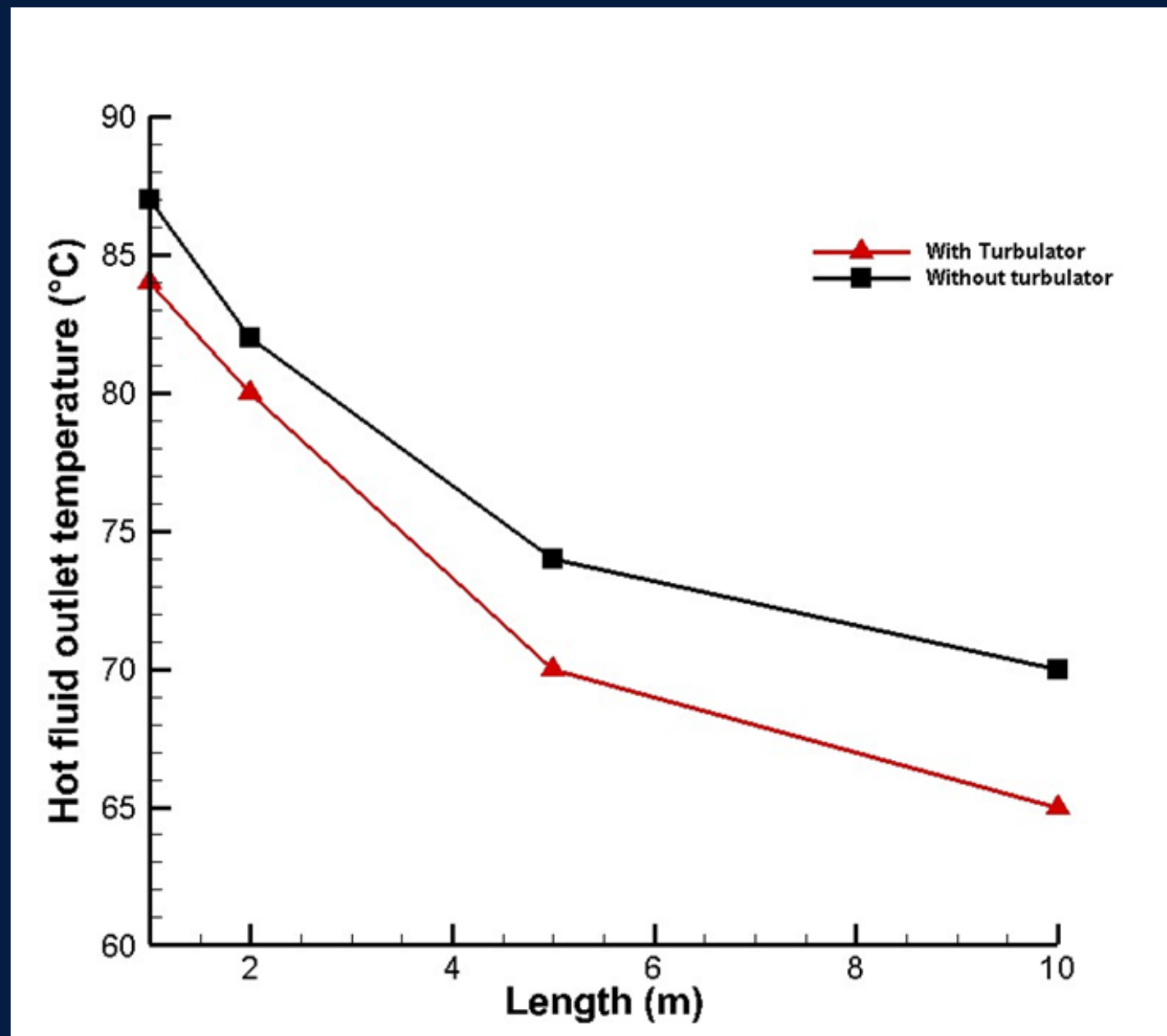
Varying mass flow rate



- The mass flow rate in the annular area (hot fluid) was changed keeping the flow rate of the internal area fixed (0.08 Kg/s).
- The cold fluid outlet temperature also increases with the mass flow rate

Parametric Analysis using CDF

Varying Length



- Analyzed lengths are 1m, 2m, 5m and 10m.
- The hot fluid outlet temperature decreases with the increase of length

Limitations of Numerical Analysis

- We were unable to calculate the effect of fin.
- The software was unable to calculate the pressure drop.
- Results were not 100% accurate due to some assumptions.

Improvements of Numerical Analysis

- Mesh size can be decreased in order to get more accurate results.
- Mesh refinement will allow us to get better results.
- Instead of using hexahedral mesh other better methods can be used.

CONCLUSION

This was a huge opportunity to think and grow as engineers. Hopefully we will be able to apply the lessons learned from this to broader engineering field.

OUR TEAM



Mohammad
Abdul
Mohaimin Rafi



Fahmid
Faiyaj
Chowdhury



Tasmila
Hossain
Sanjana



Tahsin
Haque

THANK YOU